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## **The MOF-driven synthesis of supported palladium clusters with catalytic activity for carbene-mediated chemistry**

Francisco R. Fortea-Pérez<sup>1</sup>, Marta Mon<sup>1</sup>, Jesús Ferrando-Soria<sup>1</sup>, Mercedes Boronat,<sup>2</sup>  
Antonio Leyva-Pérez<sup>2\*</sup>, Avelino Corma<sup>2\*</sup>, Juan Manuel Herrera<sup>3</sup>, Dmitrii Osadchii<sup>4</sup>,  
Jorge Gascon<sup>4</sup>, Donatella Armentano<sup>5\*</sup> and Emilio Pardo<sup>1\*</sup>

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## Supplementary Methods

**Materials.** All chemicals were of reagent grade quality. They were purchased from commercial sources and used as received. Crystals of  $\text{Mg}_2^{\text{II}}\{\text{Mg}^{\text{II}}_4[\text{Cu}^{\text{II}}_2(\text{Me}_3\text{mpba})_2]_3\} \cdot 45\text{H}_2\text{O}$  (**1**),  $\text{Ni}_2^{\text{II}}\{\text{Ni}^{\text{II}}_4[\text{Cu}^{\text{II}}_2(\text{Me}_3\text{mpba})_2]_3\} \cdot 54\text{H}_2\text{O}$  (**2**) and  $[\text{Pd}(\text{NH}_3)_4]\text{Cl}_2$  were prepared as previously reported.<sup>1</sup> Alternatively, a large scale synthesis of  $\text{Ni}_2^{\text{II}}\{\text{Ni}^{\text{II}}_4[\text{Cu}^{\text{II}}_2(\text{Me}_3\text{mpba})_2]_3\} \cdot 54\text{H}_2\text{O}$  (**2**) was carried out by direct reaction of two aqueous solutions (250 mL each) of  $\text{Na}_4[\text{Cu}_2(\text{Me}_3\text{mpba})_2] \cdot 4\text{H}_2\text{O}$  (8.71 g, 0.010 mol) and  $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$  (3.88 g, 0.013 mol) and subsequent addition, after filtration and re-suspension in 150 mL water of the resulting compound, of 1.94 g (0.0067 mol) of  $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$  (Yield 99 %).

**$[\text{Pd}^{\text{II}}(\text{NH}_3)_4][\text{Pd}^{\text{II}}_2(\mu\text{-O})(\text{NH}_3)_6(\text{NH}_4)_2]_{0.5}\{\text{Ni}^{\text{II}}_4[\text{Cu}^{\text{II}}_2(\text{Me}_3\text{mpba})_2]_3\} \cdot 52\text{H}_2\text{O}$  (**3**):** Well-formed deep green prisms of **3**, which were suitable for X-ray diffraction, were obtained by immersing crystals of **2** (*ca.* 5 mg, 0.0015 mmol) for 48 hours in 5 mL of a  $[\text{Pd}(\text{NH}_3)_4]\text{Cl}_2$  aqueous solution (0.004 mmol). Aiming at industrial applications, a multigram scale procedure was also carried out by using the same synthetic procedure but with greater amounts of both, a powder sample of compound **2** (20 g, 5.8 mmol) and  $[\text{Pd}(\text{NH}_3)_4]\text{Cl}_2$  (3.43 g, 14.0 mmol), with the same successful results and a very high yield (20.33 g, 96%). Anal.: calcd (%) for  $\text{Cu}_6\text{Ni}_4\text{Pd}_2\text{C}_{78}\text{H}_{189}\text{N}_{20}\text{O}_{88.5}$  (3652.3): C, 25.65; H, 5.22; N, 7.67. Found: C, 25.68; H, 5.13; N, 7.33. IR (KBr):  $\nu = 3014, 2951$  and  $2913\text{ cm}^{-1}$  (C–H),  $1607\text{ cm}^{-1}$  (C=O).

**$[\text{Pd}]_{0.5}@\text{Na}_3\{\text{Ni}^{\text{II}}_4[\text{Cu}^{\text{II}}_2(\text{Me}_3\text{mpba})_2]_3\} \cdot 56\text{H}_2\text{O}$  (**4**):** Both, crystals (*ca.* 5 mg) and a powder polycrystalline sample of **3** (*ca.* 10 g), were suspended in 50 mL of a  $\text{H}_2\text{O}/\text{CH}_3\text{OH}$  (1:2) solution to which an excess of  $\text{NaBH}_4$ , divided 26 in fractions (each fraction consisting of 1 mole of  $\text{NaBH}_4$  per mole of **3** to give a final  $\text{NaBH}_4$  / MOF molar ratio of 26 or, which is the same,  $\text{NaBH}_4$  / Pd atom molar ratio of 13), was added progressively in the space of 72 hours. After each addition, the mixture was allowed to react for 1.5 hour. After this period, samples were gently washed with a  $\text{H}_2\text{O}/\text{CH}_3\text{OH}$  solution and filtered on paper giving excellent yields (98%). Anal.: calcd (%) for

<sup>1</sup> (a) T. Grancha, J. Ferrando-Soria, H.-C. Zhou, J. Gascon, B. Seoane, J. Pasán, O. Fabelo, M. Julve, E. Pardo, *Angew. Chem., Int. Ed.* **2015**, *54*, 6521–6525. (b) Prepared as reported in Sintez kompleksnykh soedinenii metallov platinovoi grupy. Spravochnik (Synthesis of Complex Compounds of the Platinum Group Metals. Handbook), Chernyaev, I.I., Ed., Moscow: Nauka, 1964, p. 325.

<sup>2</sup> J. Ferrando-Soria, P. Serra-Crespo, M. de Lange, J. Gascon, F. Kapteijn, M. Julve, J. Cano, F. Lloret, J. Pasán, C. Ruiz-Pérez, et al., *J. Am. Chem. Soc.* **2012**, *134*, 15301–4.

$\text{Cu}_6\text{Ni}_4\text{Pd}_2\text{Na}_3\text{C}_{78}\text{H}_{172}\text{N}_{12}\text{O}_{92}$  (3648.1): C, 25.68; H, 4.75; N, 4.61. Found: C, 25.68; H, 4.63; N, 4.65. IR (KBr):  $\nu = 3013, 2964$  and  $2914\text{ cm}^{-1}$  (C–H),  $1603\text{ cm}^{-1}$  (C=O).

**Table 1.** Selected data from the ICP–MS<sup>a</sup>, SEM/EDX<sup>b</sup> and TEM/EDX<sup>c</sup> analyses.

| <b>Compound 3</b> |                           |                                        |                           |                                        |                           |                                        |
|-------------------|---------------------------|----------------------------------------|---------------------------|----------------------------------------|---------------------------|----------------------------------------|
| <b>Metal</b>      | <i>% mass<sup>a</sup></i> | <i>Metal stoichiometry<sup>a</sup></i> | <i>% mass<sup>b</sup></i> | <i>Metal stoichiometry<sup>b</sup></i> | <i>% mass<sup>c</sup></i> | <i>Metal stoichiometry<sup>c</sup></i> |
| <b>Cu</b>         | 10.571                    | 6.00                                   | 10.43                     | 5.92                                   | –                         | –                                      |
| <b>Ni</b>         | 6.518                     | 4.01                                   | 6.47                      | 3.98                                   | 6.46                      | 3.99                                   |
| <b>Pd</b>         | 5.892                     | 1.99                                   | 5.77                      | 1.95                                   | 5.92                      | 2.00                                   |
| <b>Compound 4</b> |                           |                                        |                           |                                        |                           |                                        |
| <b>Metal</b>      | <i>% mass<sup>a</sup></i> | <i>Metal stoichiometry<sup>a</sup></i> | <i>% mass<sup>b</sup></i> | <i>Metal stoichiometry<sup>b</sup></i> | <i>% mass<sup>c</sup></i> | <i>Metal stoichiometry<sup>c</sup></i> |
| <b>Cu</b>         | 10.802                    | 6.02                                   | 10.92                     | 6.08                                   | –                         | –                                      |
| <b>Ni</b>         | 6.679                     | 4.03                                   | 6.73                      | 4.05                                   | 6.66                      | 4.02                                   |
| <b>Pd</b>         | 6.012                     | 1.99                                   | 6.08                      | 2.02                                   | 6.07                      | 2.01                                   |
| <b>Na</b>         | 1.895                     | 2.92                                   | –                         | –                                      | –                         | –                                      |

<sup>a</sup>Solid samples were digested with 0.5 mL of HNO<sub>3</sub> 69% at 60°C for 4 hours followed by the addition of 0.5 mL of HCl 37% and digestion 80°C for 1 hour.

**X-ray crystallographic data collection and structure refinement.** Crystals of **3** and **4** with 0.12 x 0.10 x 0.10 mm as dimensions were selected and mounted on a MITIGEN holder in Paratone oil and very quickly placed on a liquid nitrogen stream cooled at 90 K to avoid the possible degradation upon dehydration. Diffraction data were collected on a Bruker–Nonius X8APEXII CCD area detector diffractometer using graphite–monochromated Mo–K $\alpha$  radiation ( $\lambda = 0.71073$  Å). The data were processed through the SAINT<sup>3</sup> reduction and SADABS<sup>4</sup> multi-scan absorption software. The structure was solved with the SHELXS structure solution program, using the Patterson method. The model was refined with version 2013/4 of SHELXL against  $F^2$  on all data by full-matrix least squares.<sup>5</sup>

As reported in the main text, the robustness of the 3D network, allowed the resolution of the crystal structure of both **3** and **4**, being their crystals suitable for X-ray diffraction, even over two- and three-step process, after a crystal-to-crystal transformation. For that reason it is reasonable to observe a quite poor diffraction power of the samples even if in presence of heavy atoms as palladium. In fact, a completeness of data was obtained at  $\theta_{\max}$  of 21 and 23°, for **3** and **4**, respectively (Table S2) (detected as Alerts A in the checkcif). However, the solution and refinement parameters are suitable, compared with MOFs structures generally reported, thus we are convinced that the structures found are consistent.

In both samples, all non-hydrogen atoms were refined anisotropically except some highly dynamically disordered atoms of the ligand [N1, C5, C6, C8, C9, C22, C23] and solvent molecules [O5W, O6W, O7W and O17W] (**3**), [O5, O6, N2, C13] and [O5W–O14W] together with Pd2, Pd3 and Pd4 palladium atoms in **4**. The occupancy factors, of Pd and Na<sup>+</sup> ions have been defined in agreement with SEM and ICP–MS results. The use of some C–C bond lengths and planes restraints on atoms that are supposed to lie on a common plane (FLAT command) during the refinements and Pd–N (**3**) or Pd–Pd (**4**) bond lengths restraints of some highly disordered atoms has been reasonable imposed and related to flexibility of the three-substituted phenyl rings of the Me<sub>3</sub>mpba ligand that are dynamic components of the frameworks. This feature is particularly marked in

<sup>3</sup> SAINT, version 6.45, Bruker Analytical X-ray Systems, Madison, WI, 2003.

<sup>4</sup> Sheldrick G.M. SADABS Program for Absorption Correction, version 2.10, Analytical X-ray Systems, Madison, WI, 2003

<sup>5</sup> (a) G. M. Sheldrick, *Acta Cryst.* **2015**, C71, 3–8. (b) G. M. Sheldrick, *Acta Cryst.* **2008**, A64, 112–122. (c) SHELXTL-2013/4, Bruker Analytical X-ray Instruments, Madison, WI, **2013**.

compound **3** whereas in **4** the further stabilization of the net by means of Na<sup>+</sup> ions (Supplementary Fig. 5) efficiently reduces such disorder. As a consequence, Alert A in the checkcifs, also related to short intra H $\cdots$ H, are detected. Furthermore, in **3**, Pd3 metal ions is disordered being placed in special position and at a very short distance to the equivalent one (levels Alert A in checkcifs).

The solvent molecules were highly disordered (some refined double positions are detected as Alerts A in the checkcif) but, even if not all the ones detected by TGA analysis, have been somehow modeled in **3** and **4**, whereas no confined NH<sub>3</sub> molecules belonging to the mononuclear complex [Pd(NH<sub>3</sub>)<sub>4</sub>]<sup>2+</sup> or NH<sub>4</sub><sup>+</sup> ions have been found from the  $\Delta F$  map, the quite large channels featured by this series of MOFs likely account for that.

As a consequence, in **3** and **4**, the contribution to the diffraction pattern from the highly disordered water and NH<sub>3</sub> molecules located in the voids was subtracted from the observed data through the SQUEEZE method, implemented in PLATON.<sup>6</sup> The hydrogen atoms of the ligand were set in calculated positions and refined as riding atoms whereas for water molecules were neither found nor calculated.

A summary of the crystallographic data and structure refinement for the two compounds is given in Table S2. CCDC reference numbers are 1517224 and 1517225 for **3**, and **4**, respectively.

Somewhat high R-values especially for **3**, (levels Alert A in checkcifs) are, most likely, mainly affected by the contribution of the highly disordered solvent to the intensities of the low angle reflections.

All attempts to increase quality of data measuring on a single crystal of **3** and **4** at CRISTAL beamline of SOLEIL synchrotron facility failed, mostly due to fast crystal deterioration under synchrotron beam radiation. Of 32265 and 18344 collected reflections for **3** and **4**, respectively, only 7837 (2209 with  $I > 2\sigma I$ ) and 3711 (1878 with  $I > 2\sigma I$ ) of them were unique.

The final geometrical calculations on free voids and the graphical manipulations were carried out with PLATON<sup>6</sup> implemented in WinGX,<sup>7</sup> and CRYSTAL MAKER<sup>8</sup> programs, respectively.

### Structural details and in depth analysis of X-ray data.

As reported in the main text, the anionic host network  $\{\text{Ni}^{\text{II}}_4[\text{Cu}^{\text{II}}_2(\text{Me}_3\text{mpba})_2]_3\}^{4-}$  displays remarkably robust crystallinity, which enables the use of X-ray crystallography, as ultimate method of structural determination, for the unambiguously *in situ* observation of guest inclusion, either after absorption of  $[\text{Pd}^{\text{II}}(\text{NH}_3)]_4\text{Cl}_2$  guest molecules (**3**) or after the reduction process (**4**). Both crystal structures **3** and **4** clearly evidence the presence of  $[\text{Pd}(\text{NH}_3)_4]^{2+}$  or  $[\text{Pd}^{\text{II}}_2(\mu\text{-O})(\text{NH}_3)_6]^{2+}$  and  $\text{Pd}_4$  NCs guest hosted in the nanopores of **3** and **4**.

The anionic  $\text{Ni}^{\text{II}}_4\text{Cu}^{\text{II}}_6$  open-framework structure, in **3** and **4**, exhibit a pillared square/octagonal layer architecture, where nickel(II) and copper(II) ions are located on the vertices and midpoints of the edges, respectively, featuring three types of pores, different in size and shape, propagating along the *c* axis. In fact, it is built up of regularly spaced, almost square sized small pores (virtual diameter of *ca.* 0.4 nm) and two octagonal-type large pores (virtual diameters of *ca.* 1.5 and 2.2 nm, respectively), hydrophobic and hydrophilic, depending on the disposition of the trimethyl-substituted phenylene spacers, pointing inwards or outwards of the voids, respectively (Fig. 1). The host framework, occupies only 35% of the crystal volume, (calculated subtracting also the volume related to metal counterions allocated in the pores). As stated in the main text the arrangement of  $[\text{Pd}^{\text{II}}(\text{NH}_3)]_4^{2+}$  moieties or dinuclear complexes of the type  $[\text{Pd}^{\text{II}}_2(\mu\text{-O})(\text{NH}_3)_6]^{2+}$  in **3** and  $\text{Pd}$  NCs in **4**, confined into the channels are, as expected, strictly related.

In **3**, three crystallographically independent Pd(II) metal ions are present in the asymmetric unit whereas in **4** there are four crystallographically independent Pd(I)/Pd(0) metal ions for which positions have been refined, with occupancy factors, in agreement with SEM results. While in **3**,  $[\text{Pd}^{\text{II}}_2(\mu\text{-O})(\text{NH}_3)_6]^{2+}$  dimers interact with the net through  $\text{-NH}_3\cdots\text{O}$  hydrogen bonds (Supplementary Fig. 6), in **4** the  $\text{Pd}_4$  moieties

<sup>6</sup> Spek, A. L. *Acta Crystallogr. Sect. D, Biol. Crystallogr.* **2009**, 65, 148.

<sup>7</sup> Farrugia, L. J. *J. Appl. Crystallogr.* **1999**, 32, 837.

<sup>8</sup> D. Palmer, CRYSTAL MAKER, Cambridge University Technical Services, C. No Title, 1996.

are situated at average Pd $\cdots$ O distances of 2.8 Å far from oxygen donor atoms of the wall of the octagonal pore (Fig. 4 and Supplementary Figs 2 and 4). Hence, it is also reasonable to suppose either [Pd<sub>4</sub>]<sup>2+</sup> clusters straight stabilized by the network by means of these weak supramolecular interactions between Pd and oxamate moieties or that the presence of a hydrated primary coordination sphere around Pd can further drive interactions with the net. In fact, being highly porous, MOFs can adsorb and retain a very large amount of solvent molecules in their pores. According to this point, this feature could play a role in the stabilization of Pd<sub>4</sub> clusters in such vastly solvated nano-confined space. Lattice water molecules hydrogen-bonded to the network can therefore bind Pd clusters supporting and further stabilizing them. Looking at X-ray data, even taking into account the disorder that always affects solvent molecules and the related diffuse electron density (very difficult to model), it is possible to identify the first two peaks of residual density [1.29 and 1.24 e.Å<sup>-3</sup>] which may correspond to two coordinated waters, being at a distance of 1.9 [at Pd(4)] and 1.8 Å [Pd(2)] and thus, coordinated to Pd. Anyway, aiming at being rigorous, we prefer to not explicitly assign these residual peaks that, due to the quite large pores hosting them, give us a case of continuous positional disorder, in which any rotational angle of the molecules or part of them (oxygen atoms) are more or less energetically similar. However, we are confident that this feature can represent a clear insight about this possibility, considering a synergic effect between network and solvent molecules surrounding clusters, this stabilization might reflect the unique condition of the Pd<sub>4</sub> unit in the MOF. Continuing our analysis of X-ray data we can also highlight that Pd<sub>4</sub> clusters exhibit distances from some copper coordinated or hydrogen bonded to the net water molecules [that we were able to assign] ranging from 4.8 to 5.0 Å, in perfect agreement with a plausible primary sphere of water molecules coordinated to Pd further interacting with the net.

It is also evident from X-ray data that the positions of Pd atoms are disordered. In fact, Pd atoms in **4** were solved with considerably large thermal ellipsoids, however, the expected thermal disorder (dynamical disorder, a real motion in the solid state) cannot be held responsible for a so much shorter Pd-O<sub>oxamate</sub> distance than the detected one when considering the accuracy of X-rays [bond precision on C-C = 0.0208 Å] to envisage a more strong Pd $\cdots$ O interaction to realize an actual Pd-O bond. Even considering Pd $\cdots$ O distances [average Pd $\cdots$ O distances 2.8(1) Å] ranging within +/-

three times standard deviations, these could reach a minimum value of 2.5 Å that is still out of the range for a Pd primary sphere of coordination.

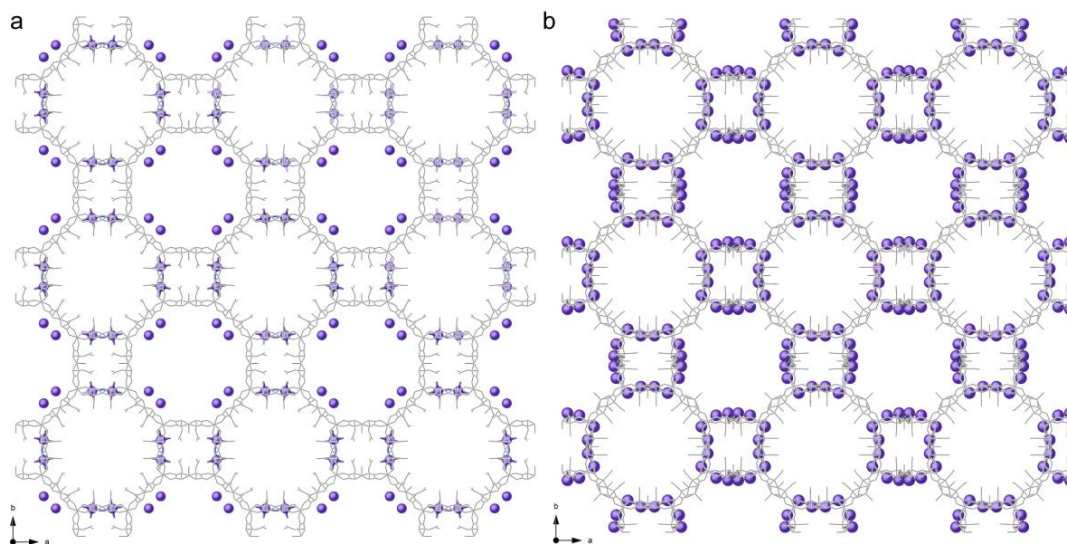
For the sake of rigour and as possible X-ray error analysis we have also to underline that, in principle, disordered [Pd<sub>2</sub>] instead of [Pd<sub>4</sub>] clusters might be consistent. In fact, we took that possibility into account when refining data. However all the factors described as follow determine that Pd<sub>4</sub> clusters are, by far, the most probable resolution. When we solved the crystal structure of **4** considering the case of disordered [Pd<sub>2</sub>] we reached refinement results not so divergent in term of goodness of fit, which means that this refinement, even if slightly worse than considering [Pd<sub>4</sub>], could be acceptable. However, considering how challenging the structural characterisation of this kind of hybrid material is, several considerations should be taken into account to drive towards the most probably results: in order to define [Pd<sub>2</sub>] clusters, we should assume that two spatial orientations of such dimers, related by a symmetry operation, on crystalline sites are allowed. This positional disorder in **4** should be considered as a static disorder distributed among different unit cells (a sort of a look-alike motion) in which the clusters can possess two well defined energetically similar positions. In fact the final positions of [Pd<sub>2</sub>] will be mediated in overall crystal assuming a 1:1 statistical distribution of the two defined positions giving as result that in cells they reside only on one side of the vector Ni···Cu···Ni of the net. This suggest us a less net-stabilized state (as confirmed by theoretical calculations) also taking into account the overall +2 charge of the cluster [Pd<sub>2</sub>]<sup>2+</sup> disordered or [Pd<sub>4</sub>]<sup>2+</sup>. Moreover, considering the multi technique approach that pervades the whole paper, the assumption of dimers would involve a random distribution of charge with most probably Pd(I)-Pd(0) together with either clusters containing only Pd(0) or Pd(I) metal/ions to be in agreement with DRX, ICP-MS, XPS results. This situation is unlikely as it is hard to envisage Pd clusters in different oxidation states where no change occur in bond lengths along dimers. It is very difficult to assume that a Pd(I)-Pd(I) bond length will be exactly the same to a Pd(0)-Pd(0) ones giving dimers possessing two energetically very similar state and thus positions.

Finally, theoretical calculations were performed to support the [Pd<sub>4</sub>]<sup>2+</sup> XRD structure (see computational details section).

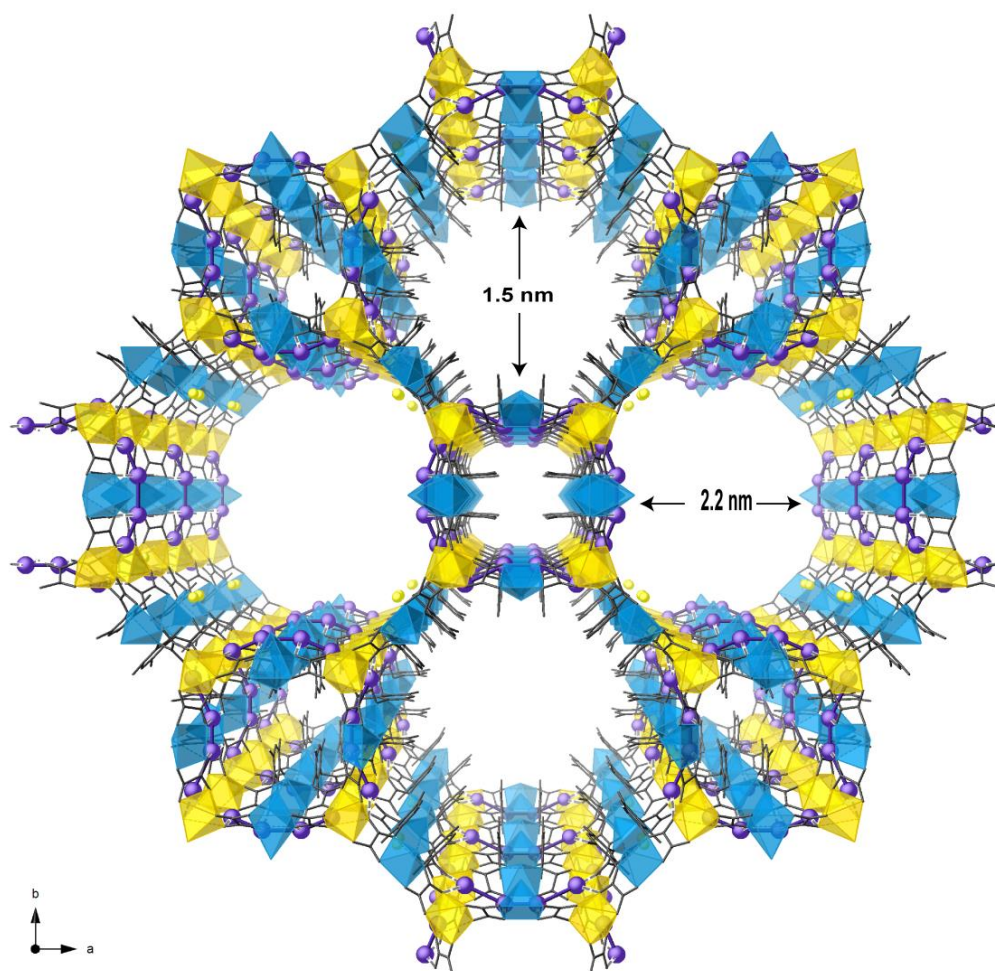
**Table 2. Summary of Crystallographic Data for 3 and 4.**

| Compound                                   | 3                                                                                                                  | 4                                                                                                                                |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Formula                                    | C <sub>78</sub> H <sub>189</sub> Cu <sub>6</sub> N <sub>20</sub> Ni <sub>4</sub> O <sub>88.5</sub> Pd <sub>2</sub> | C <sub>78</sub> H <sub>172</sub> Cu <sub>6</sub> N <sub>12</sub> Na <sub>3</sub> Ni <sub>4</sub> O <sub>92</sub> Pd <sub>2</sub> |
| <i>M</i> (g mol <sup>-1</sup> )            | 3652.35                                                                                                            | 3648.12                                                                                                                          |
| $\lambda$ (Å)                              | 0.71073                                                                                                            | 0.71073                                                                                                                          |
| Crystal system                             | Tetragonal                                                                                                         | Tetragonal                                                                                                                       |
| Space group                                | <i>P4/mmm</i>                                                                                                      | <i>P4/mmm</i>                                                                                                                    |
| <i>a</i> (Å)                               | 35.920(2)                                                                                                          | 35.794(13)                                                                                                                       |
| <i>c</i> (Å)                               | 15.3561(9)                                                                                                         | 15.063(5)                                                                                                                        |
| <i>V</i> (Å <sup>3</sup> )                 | 19813(2)                                                                                                           | 19298(16)                                                                                                                        |
| <i>Z</i>                                   | 4                                                                                                                  | 4                                                                                                                                |
| $\rho_{\text{calc}}$ (g cm <sup>-3</sup> ) | 1.224                                                                                                              | 1.256                                                                                                                            |
| $\mu$ (mm <sup>-1</sup> )                  | 1.259                                                                                                              | 1.299                                                                                                                            |
| <i>T</i> (K)                               | 90                                                                                                                 | 90                                                                                                                               |
| $\theta$ range for data collection (°)     | 0.80 to 23.24                                                                                                      | 0.80 to 20.87                                                                                                                    |
| Completeness to $\theta = 23.0^\circ$      | 100%                                                                                                               | $\theta = 21.0^\circ$ 99.5%                                                                                                      |
| Completeness to $\theta = 25.0$            | 80%                                                                                                                | 59%                                                                                                                              |
| Measured reflections                       | 96726                                                                                                              | 56926                                                                                                                            |
| Unique reflections (Rint)                  | 7880(0.0443)                                                                                                       | 5693(0.1162)                                                                                                                     |
| Observed reflections [ $I > 2\sigma(I)$ ]  | 6354                                                                                                               | 3109                                                                                                                             |
| Goof                                       | 2.171                                                                                                              | 1.137                                                                                                                            |
| $R^a$ [ $I > 2\sigma(I)$ ] (all data)      | 0.2275(0.2442)                                                                                                     | 0.1206(0.1808)                                                                                                                   |
| $wR^b$ [ $I > 2\sigma(I)$ ] (all data)     | 0.5840(0.5964)                                                                                                     | 0.3279(0.3571)                                                                                                                   |

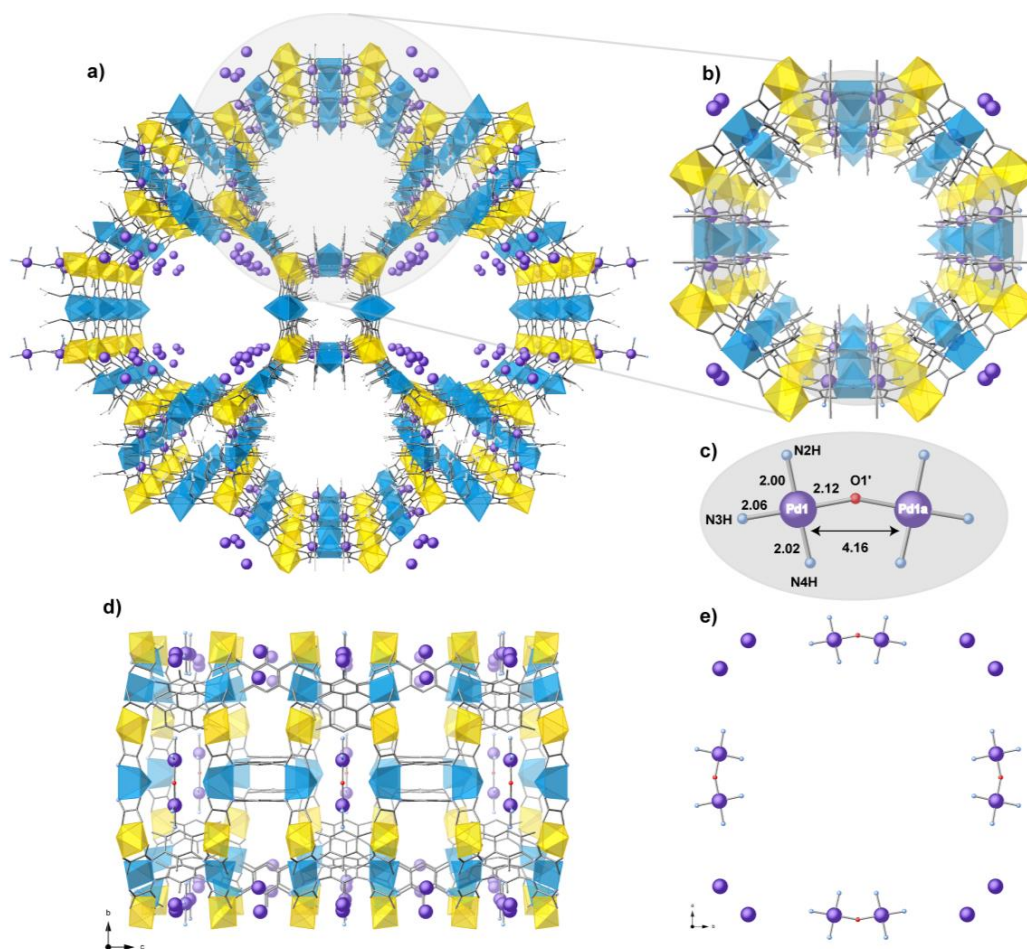
<sup>a</sup>  $R = \sum(|F_o| - |F_c|) / \sum |F_o|$ . <sup>b</sup>  $wR = [\sum w(|F_o| - |F_c|)^2 / \sum w|F_o|^2]^{1/2}$ .



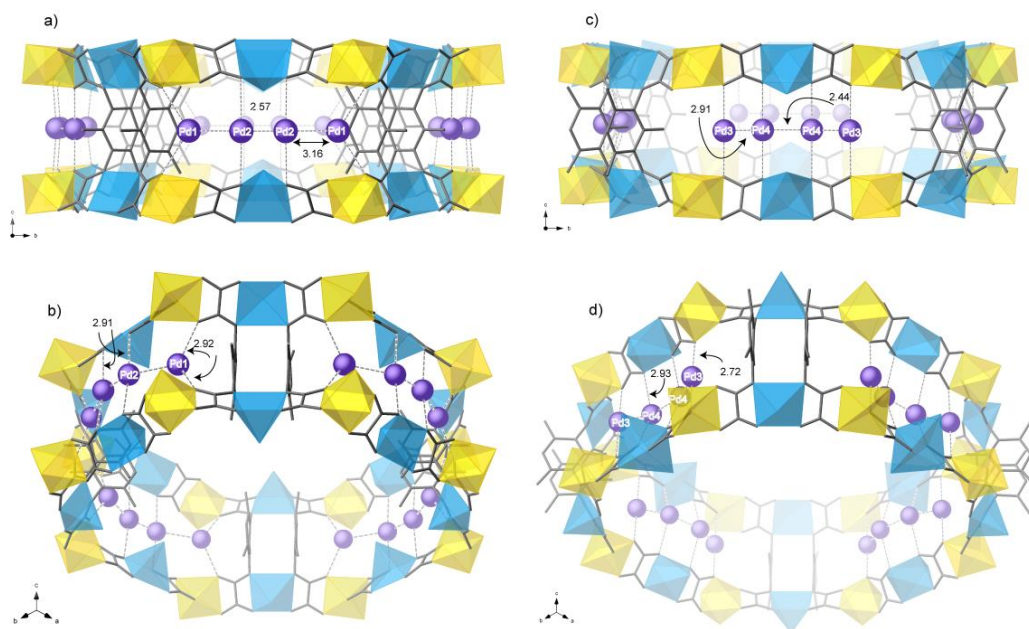
**Supplementary Figure 1.** Perspective view along *c* crystallographic axis of crystal structures of **3** (a) and **4** (b) featuring channels filled by Pd(II) complexes (**3**) or [Pd<sub>4</sub>]<sup>2+</sup> NCs (**4**). Lattice water molecules and hydrogen atoms have been omitted for clarity. Color scheme: palladium, violet sphere; ligands atoms and metal ions of the whole net have been depicted as grey sticks.



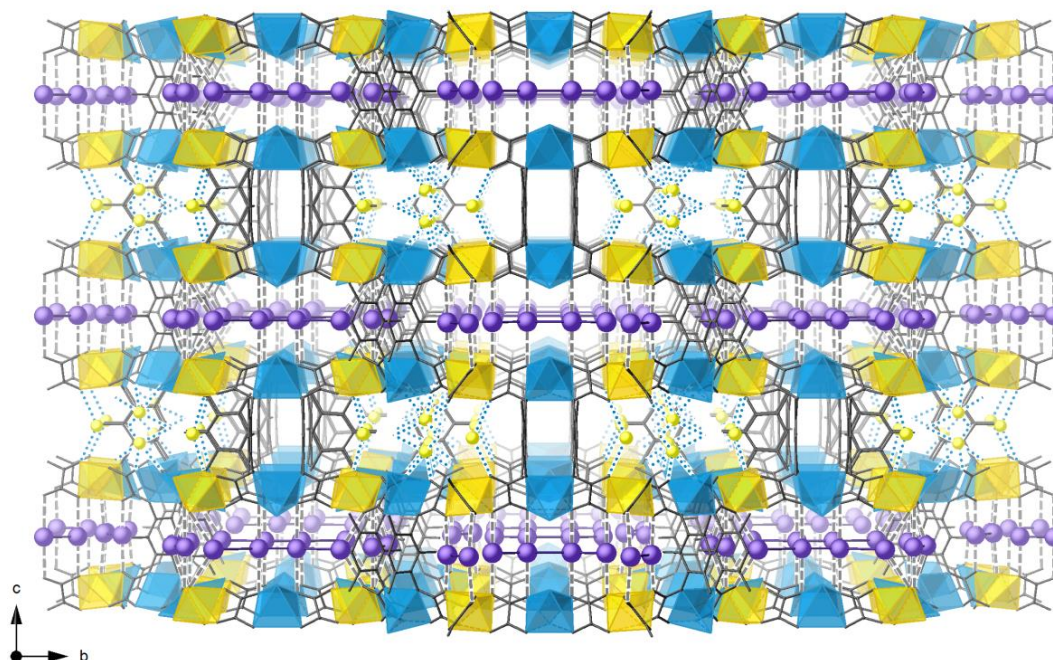
**Supplementary Figure 2.** Perspective view along *c* crystallographic axis of crystal structure of **4** featuring hydrophobic and hydrophilic octagonal pores, exhibiting a virtual diameter of 1.5 and 2.2 nm, respectively. the hydrophobic ones accommodate  $[\text{Pd}_4]^{2+}$  NCs and counterbalancing alkali Na(I) cations as a result of the reduction with  $\text{NaBH}_4$ . Lattice water molecules and hydrogen atoms have been omitted for clarity. Color scheme: Sodium(I), yellow spheres; copper(II), blue polyhedra; nickel(II), yellow gold polyhedra; palladium, violet spheres. Ligands atoms have been depicted as grey sticks.



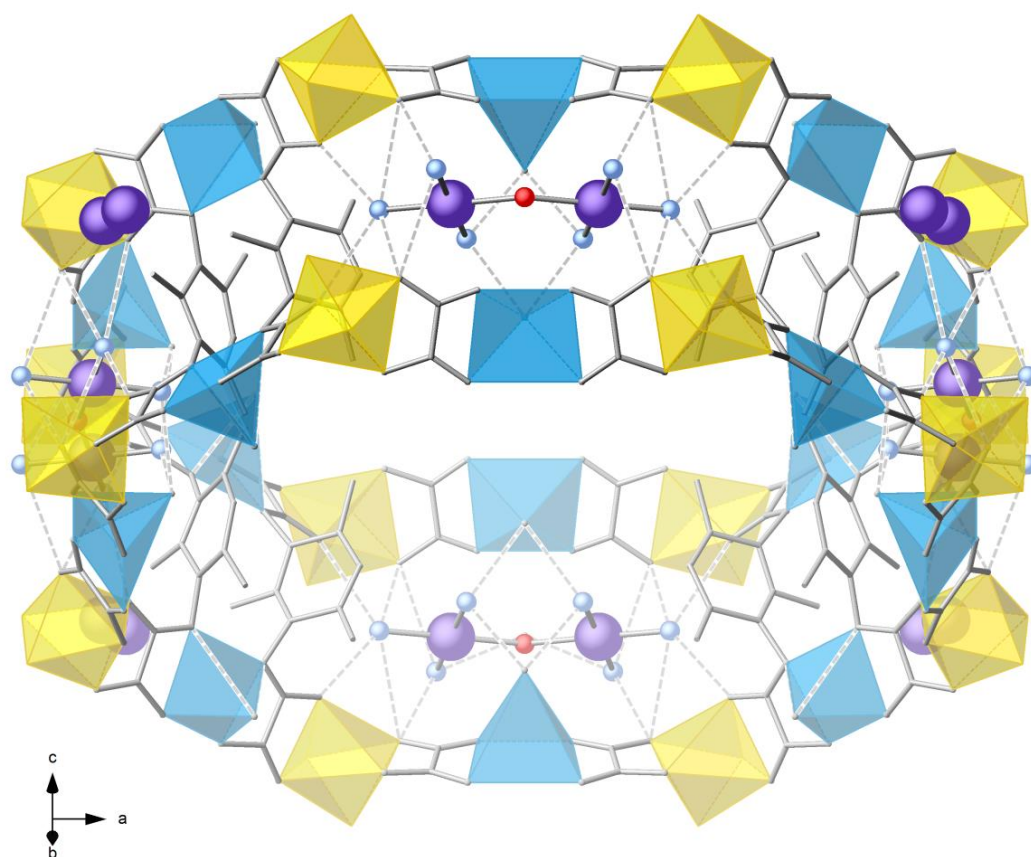
**Supplementary Figure 3.** Details of crystal structure of **3**: a) Perspective view along *c* crystallographic axis of a portion of the highly porous net where mononuclear Pd(II) solvated and dinuclear  $[\text{Pd}^{\text{II}}_2(\mu\text{-O})(\text{NH}_3)_6]^{2+}$  complexes, formed after the insertion of  $[\text{Pd}^{\text{II}}(\text{NH}_3)_4]^{2+}$  by a dimerization reaction, reside; details along *c* axis of b) a pore with intercalated Pd(II) ions and complexes and of c) the, surprisingly, highly ordered structure of  $[\text{Pd}^{\text{II}}_2(\mu\text{-O})(\text{NH}_3)_6]^{2+}$  and its structural parameters; d) perspective view of a channel along *a* axis; e) view along *c* axis of Pd(II) ions and complexes in **3**. Lattice water molecules and hydrogen atoms have been omitted for clarity. Color scheme: Sodium(I), yellow spheres; copper(II), blue polyhedra; nickel(II), yellow gold polyhedra; palladium, violet spheres; nitrogen from  $\text{NH}_3$ , light blue spheres;  $\text{O}^{2-}$ , red spheres; ligand atoms have been depicted as grey sticks.



**Supplementary Figure 4.** A portion of a pore of the crystal structure of **4** along the *a* axis (a and b) and the [111] direction (c and d) showing the two not-crystallographically equivalent tetranuclear Pd<sub>4</sub> NCs of the type Pd<sup>+</sup>Pd<sup>0</sup>Pd<sup>0</sup>Pd<sup>+</sup> and related structural parameters, stabilized by Pd···O interactions [distances are in Å]. Lattice water molecules and hydrogen atoms have been omitted for clarity. Color scheme: Sodium(I), yellow spheres; copper(II), blue polyhedra; nickel(II), yellow gold polyhedra; palladium, violet spheres. Ligands atoms have been depicted as grey sticks.



**Supplementary Figure 5.** Perspective view of crystal structure of **4**, showing hydrated charge-counterbalancing alkali  $\text{Na}^{\text{I}}$  cations distributed within the channels, further stabilizing the whole net, through interactions with the carboxylate- and/or carbonyl-oxygen atoms from the coordination network (blue dotted lines) occupying the interlayer space of the wide octagonal channels. Color scheme: Sodium(I), yellow spheres; copper(II), blue polyhedra; nickel(II), yellow gold polyhedra; palladium, violet spheres. Ligands atoms have been depicted as grey sticks.



**Supplementary Figure 6.** Perspective view of a pore of the crystal structure of **3** showing  $[\text{Pd}^{\text{II}}_2(\mu\text{-O})(\text{NH}_3)_6]^{2+}$  dimers retained by the net through  $\text{-NH}_3\cdots\text{O}$  hydrogen bonds [ $\text{NH}_3\cdots\text{O}$  distances vary in the range 2.91–3.28 Å]. Lattice water molecules and hydrogen atoms have been omitted for clarity. Color scheme: Sodium(I), yellow spheres; copper(II), blue polyhedra; nickel(II), yellow gold polyhedra; palladium, violet spheres; nitrogen from  $\text{NH}_3$ , light blue spheres;  $\text{O}^{2-}$ , red spheres; ligand atoms have been depicted as grey sticks.

**Computational details.** All calculations are based on Density Functional Theory (DFT) and were carried out with the Gaussian 09 program package,<sup>9</sup> using the hybrid B3PW91 functional, which combines the PW91 correlation functional by Perdew and Wang with the Becke's hybrid three-parameter exchange functional.<sup>10</sup> The LANL2DZ basis set that combines Los Alamos effective core potential with a double zeta (DZ) set for valence electrons was used for Pd, Cu and Ni atoms,<sup>11</sup> while the standard 6-311G(d,p) basis set by Pople was employed for O, N, C and H atoms.<sup>12</sup> Atomic charges were calculated using the natural bond order (NBO) approach.<sup>13</sup>

A cluster model containing two Cu<sup>2+</sup> cations, four Ni<sup>2+</sup> cations and six Me<sub>3</sub>mpba units, {Ni<sub>4</sub>[Cu<sub>2</sub>(Me<sub>3</sub>mpba)<sub>2</sub>]<sub>3</sub>}<sup>4+</sup> was initially cut out from the experimental geometry of material **4**, and further simplified by substituting four aromatic rings of the Me<sub>3</sub>mpba units by methyl groups. To maintain in this cluster model the same charge balance as in the global system, two of the four N atoms that in the real system would be interacting with Cu<sup>2+</sup> cations in the next coordination sphere were saturated with H atoms. The resulting system, Cu<sub>2</sub>Ni<sub>4</sub>N<sub>8</sub>O<sub>24</sub>C<sub>38</sub>H<sub>34</sub><sup>2-</sup> depicted in Supplementary Fig. 7, includes all atoms that could stabilize the Pd<sub>4</sub> clusters by means of short and medium range interactions. Then, four Pd atoms were placed in the centre of the octagonal pore as indicated by the XRD analysis, and their positions were allowed to fully optimize without restrictions while keeping the rest of the model fixed.

**Physical Techniques.** Elemental (C, H, N) analyses were performed at the Microanalytical Service of the Universitat de València. <sup>1</sup>H, <sup>13</sup>C and DEPT NMR spectra were recorded at room temperature on a Bruker AC 200 (200.1 MHz) spectrometer. FT-IR spectra were recorded on a Perkin-Elmer 882 spectrophotometer as KBr pellets. The thermogravimetric analysis was performed on crystalline samples under a dry N<sub>2</sub>

<sup>9</sup> Gaussian 09, Revision C.01, Frisch, M. J.; Trucks, G. W.; Schlegel, H. B.; Scuseria, G. E.; Robb, M. A.; Cheeseman, J. R.; Scalmani, G.; Barone, V.; Mennucci, B.; Petersson, G. A.; et al. Gaussian, Inc., Wallingford CT, 2009.

<sup>10</sup> (a) Perdew, J. P.; Chevary, J. A.; Vosko, S. H.; Jackson, K. A.; Pederson, M. R.; Singh, D. J.; Fiolhais, C. *Phys. Rev. B* **1992**, *46*, 6671–6687. (b) Perdew, J. P.; Wang, Y. *Phys. Rev. B* **1992**, *45*, 13244–13249.

(c) Becke, A. D. *J. Chem. Phys.* **1993**, *98*, 5648–5652.

<sup>11</sup> (a) Hay, P. J.; Wadt, W. R. Ab Initio Effective Core Potentials for Molecular Calculations - Potentials for the Transition-Metal Atoms Sc to Hg. *J. Chem. Phys.* **1985**, *82*, 270–283. (b) Hay, P. J.; Wadt, W. R. Ab Initio Effective Core Potentials for Molecular Calculations - Potentials for K to Au including the Outermost Core Orbitals. *J. Chem. Phys.* **1985**, *82*, 299–310.

<sup>12</sup> K. Raghavachari, J. S. Binkley, R. Seeger, and J. A. Pople, "Self-Consistent Molecular Orbital Methods. 20. Basis set for correlated wave-functions," *J. Chem. Phys.*, **72** (1980) 650–54.

<sup>13</sup> Reed, A. E.; Weinstock, R. B.; Weinhold, F. *J. Chem. Phys.* **1985**, *83*, 735–747.

atmosphere with a Mettler Toledo TGA/STDA 851<sup>e</sup> thermobalance operating at a heating rate of 10 °C min<sup>-1</sup>. Absorption spectra were recorded on an UV/Vis spectrophotometer (UV0811M209, Varian).

The N<sub>2</sub> adsorption–desorption isotherms at 77 K were carried out on crystalline samples of **3** and **4** with a Micromeritics ASAP2020 instrument. Samples were evacuated at 50 °C during 15 hours under 10<sup>-6</sup> Torr prior to their analysis.

**X-ray Powder Diffraction Measurements.** Polycrystalline samples of **3** and **4**, before and after catalysis, were introduced into 0.5 mm borosilicate capillaries prior to being mounted and aligned on a Empyrean PANalytical powder diffractometer, using Cu K $\alpha$  radiation ( $\lambda = 1.54056$  Å). For each sample, five repeated measurements were collected at room temperature ( $2\theta = 2$ –60°) and merged in a single diffractogram.

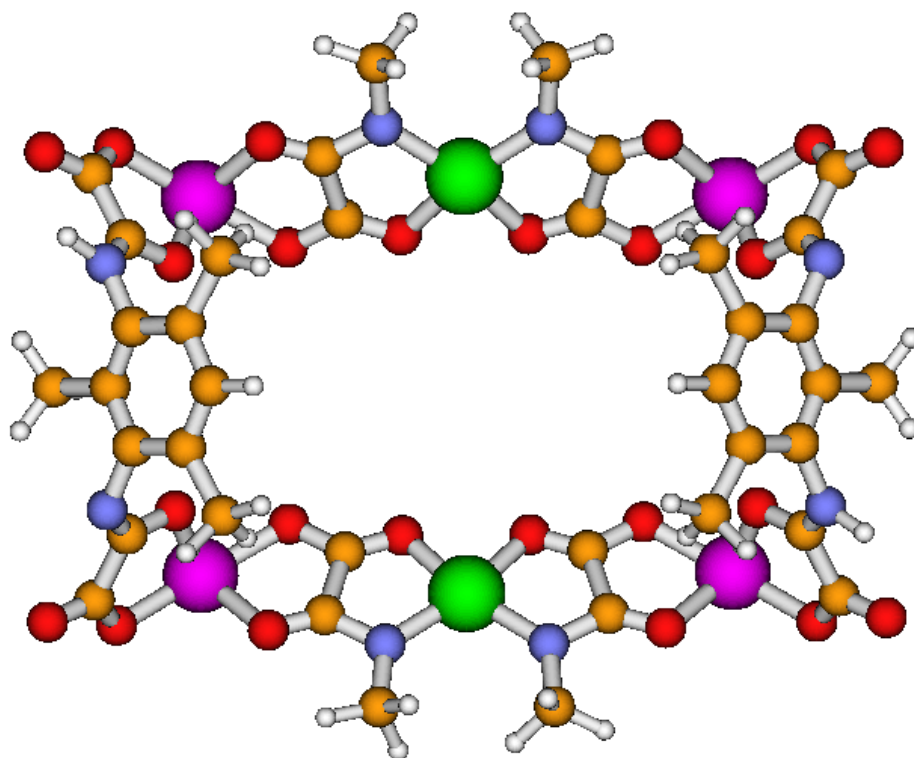
**X-ray photoelectron spectroscopy (XPS) measurements.** Samples were prepared by dropping a solid water suspension onto a molybdenum plate followed by air drying, and then measurements were performed on a SPECS spectrometer equipped with a Phoibos 150 MCD–9 analyzer using non-monochromatic Mg KR (1253.6 eV) X-ray source working at 50 W. As an internal reference for the peak positions in the XPS spectra, the C1s peak has been set at 284.5 eV.

**FTIR spectroscopy of adsorbed CO.** Fourier transform infrared (FTIR) using CO as a probe molecule was used to evaluate electronic properties of Pd<sub>4</sub>–MOF. The spectra were recorded on a Biorad FTS–40A spectrometer equipped with a DTGS detector. The experiments have been carried out in a homemade IR cell able to work in the high and low (77 K) temperature range. Prior to CO adsorption experiments, the sample was evacuated at 298 K under vacuum (10<sup>-6</sup> mbar) for 1 h. CO adsorption experiments were performed at 77 K in the 0.2–20 mbar range. Spectra were recorded once complete coverage of CO at the specified CO partial pressure was achieved. Deconvolution of the IR spectra has been performed in the Origin software using Gaussian curves where the full width at half–maximum (fwhm) of the individual bands has been taken as constant. The peak areas are normalized to the sample weight.

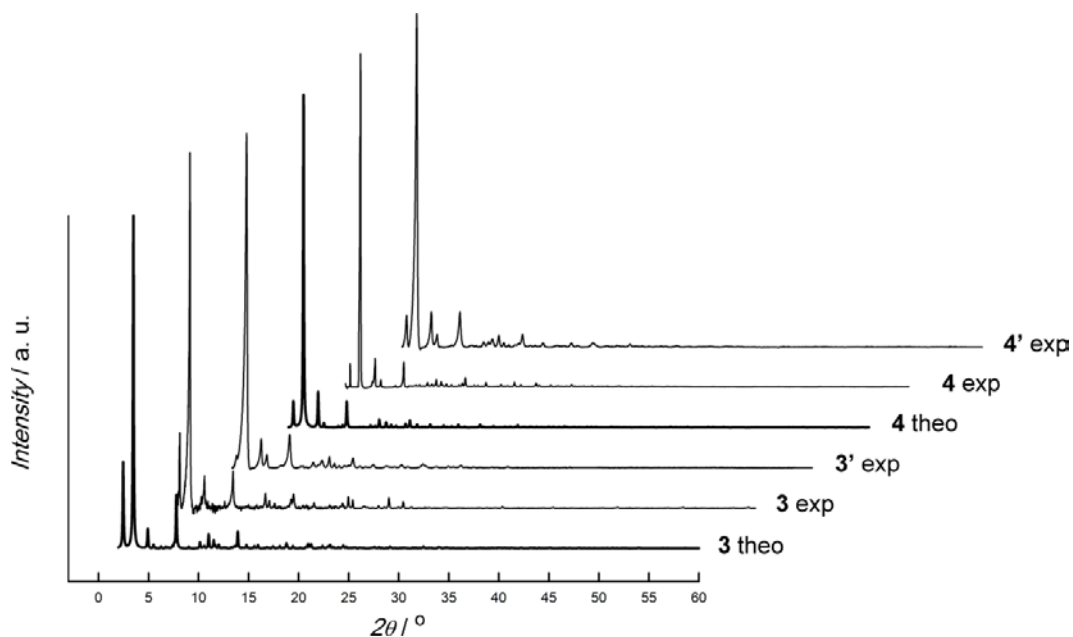
**Microscopy measurements.** High–Resolution Transmission Electron Microscopy (HR–TEM), High–Angle Annular Dark–Field Scanning Transmission Electron

microscopy (HAADF–STEM) and Energy Dispersive X–Ray Analysis (EDX) characterizations were done using a HAADF–FEI–TITAN G2 electron microscope. 5 mg of the material was redispersed in 1 mL of absolute EtOH. Carbon reinforced copper grids (200 meshes) were submerged into the suspension 30 times and then allowed to dry on air for 24 h.

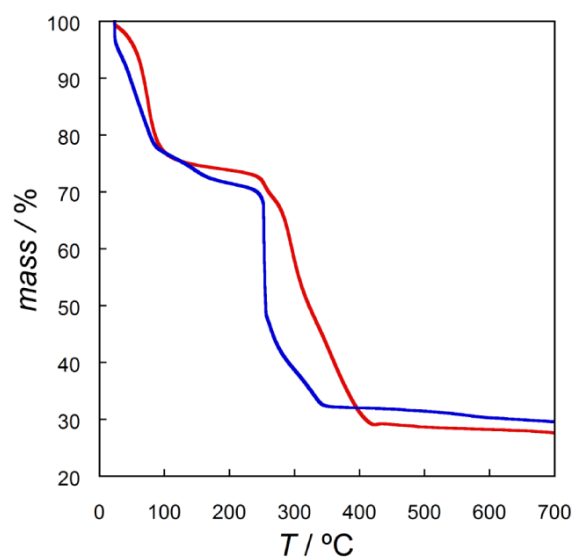
Scanning Electron Microscopy coupled with Energy Dispersive X–ray (SEM/EDX) was carried out with a XL 30 ESEM (PHILIPS) microscope equipped with a home–made EDX energy dispersive x–ray detector.



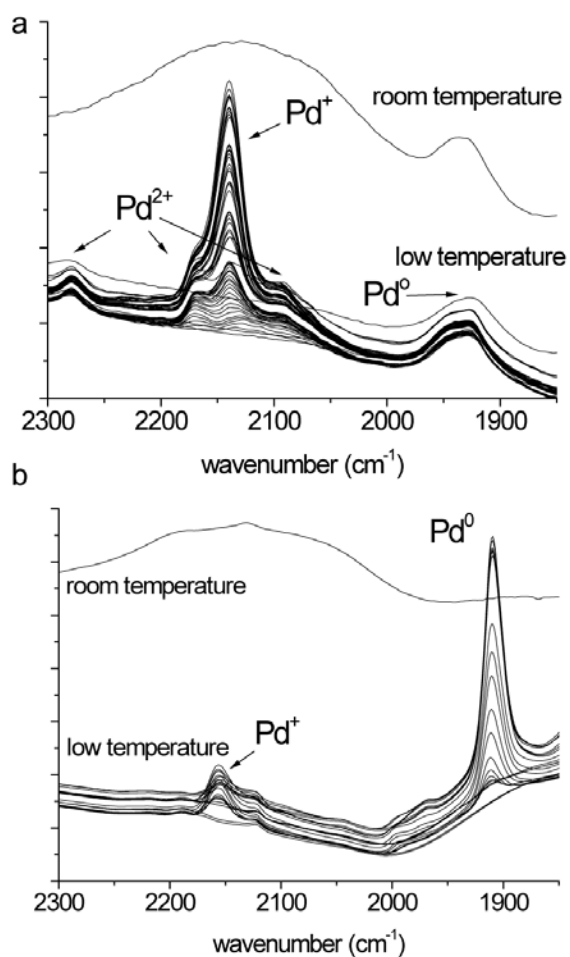
**Supplementary Figure 7.**  $\text{Cu}_2\text{Ni}_4\text{N}_8\text{O}_{24}\text{C}_{38}\text{H}_{34}^{2-}$  cluster model used in the DFT study. Cu, Ni, N, O, C and H atoms are depicted in green, pink, purple, red, orange and white, respectively.



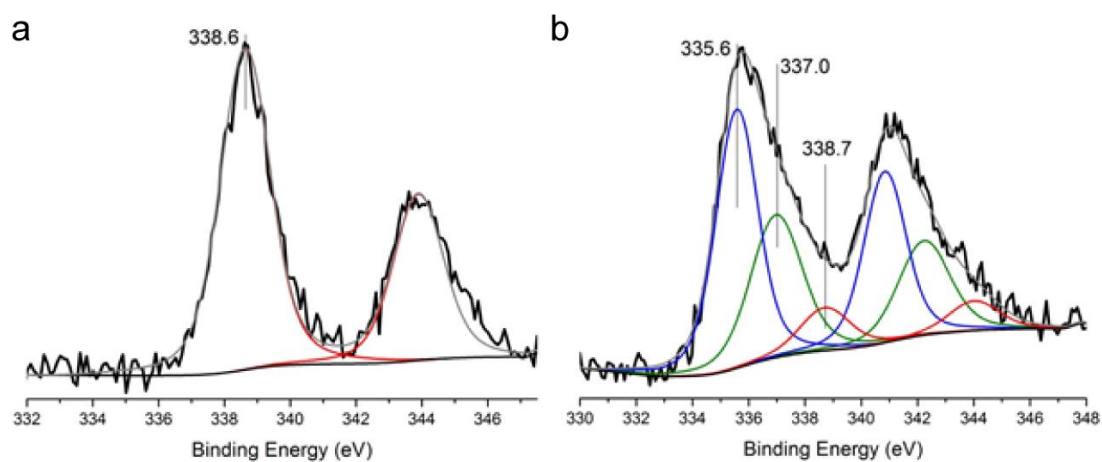
**Supplementary Figure 8.** Calculated (bold lines) and experimental (solid lines) PXRD pattern profiles of **3** and **4** in the  $2\theta$  range 2.0–60.0°. Experimental PXRD patterns are given before (**3** exp. and **4** exp.) and after (**3'** exp. and **4'** exp.) 5 catalytic cycles.



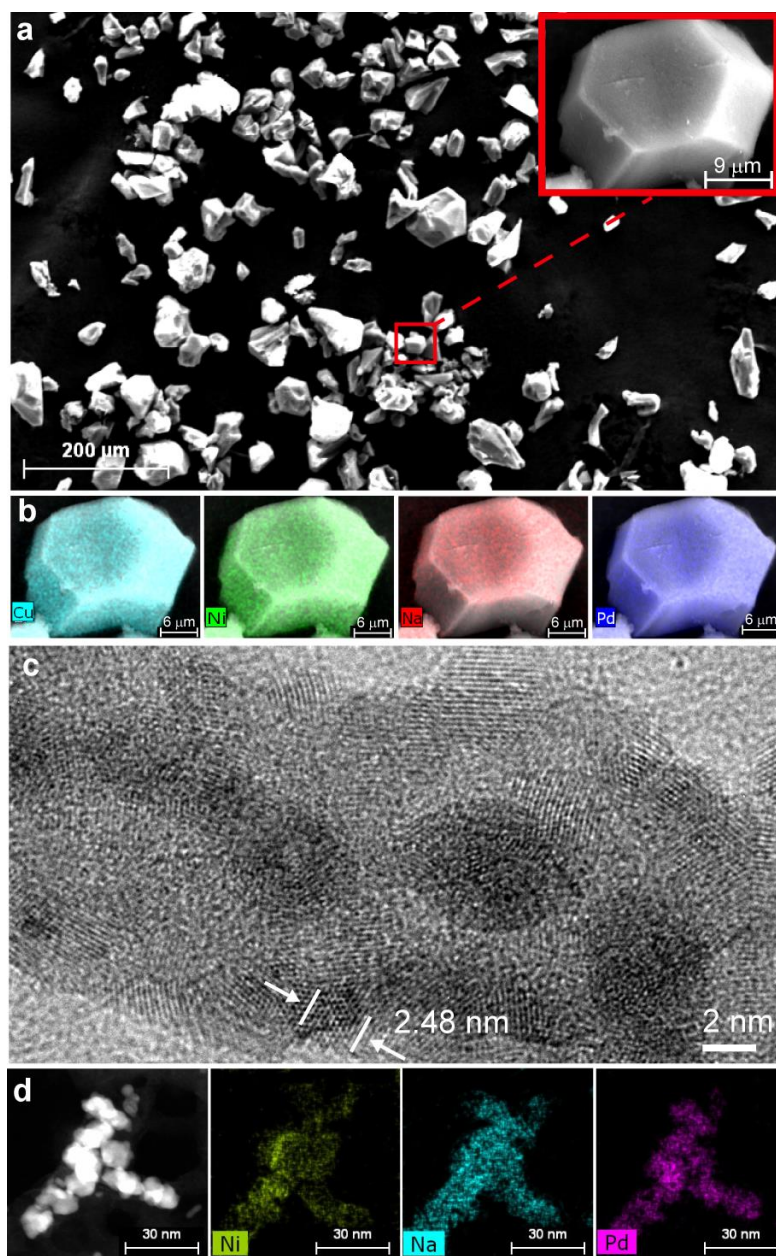
**Supplementary Figure 9.** Thermo-Gravimetric Analyses (TGA) of **3** (red) and **4** (blue) under dry N<sub>2</sub> atmosphere.



**Supplementary Figure 10.** Fourier-transformed infrared spectrum (FTIR) of adsorbed CO at room temperature and also at low temperature ( $-175\text{ }^{\circ}\text{C}$ ) on **4** (a) and, for comparison, FTIR spectrum of Pd supported and reduced on zeolite LH (b), recorded under identical conditions.



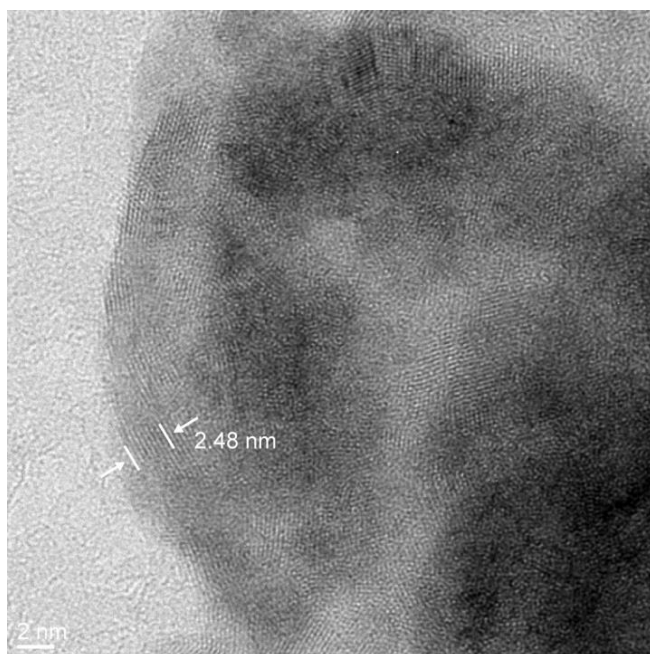
**Supplementary Figure 11.** X-ray photoelectron spectroscopy (XPS) of **3** (a) and **4** (b).



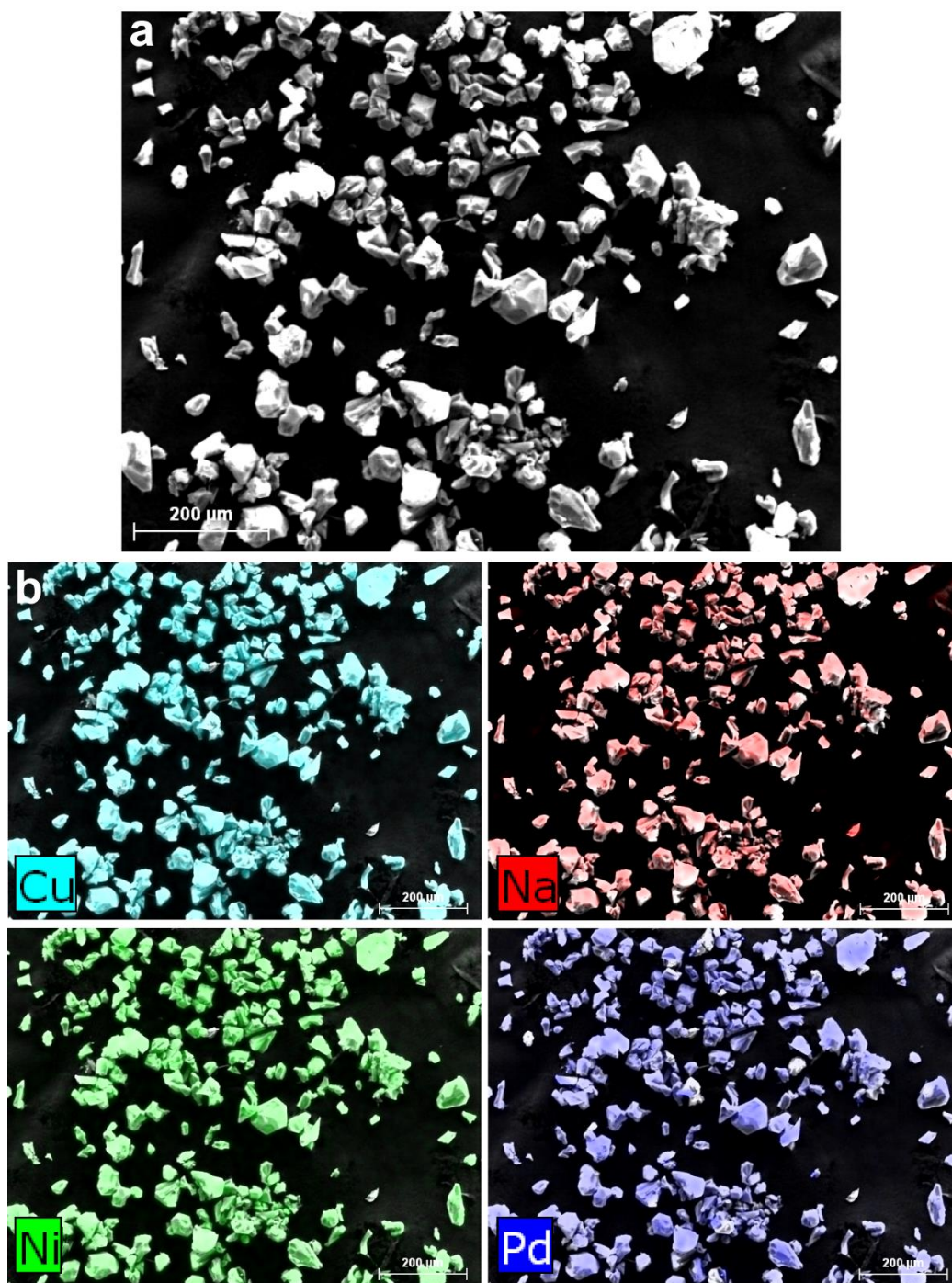
**Supplementary Figure 12.** **a.** SEM image of a polycrystalline sample of **4** and the corresponding EDX elemental mapping of one selected crystal for Cu, Ni, Pd, and Na elements (**b**). **c.** HR-TEM image of **4**. **d.** HAADF-STEM image of **4** and the corresponding EDX elemental mapping for Ni, Pd, and Na elements.



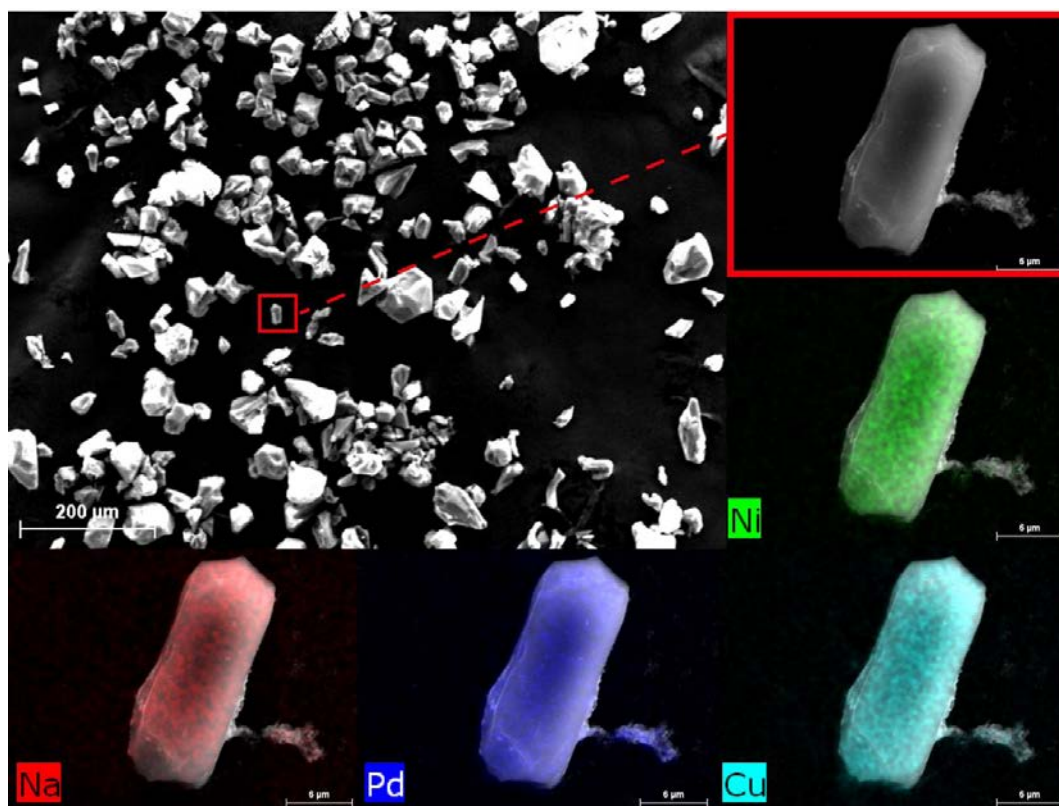
**Supplementary Figure 13.** HR-TEM image of **4**.



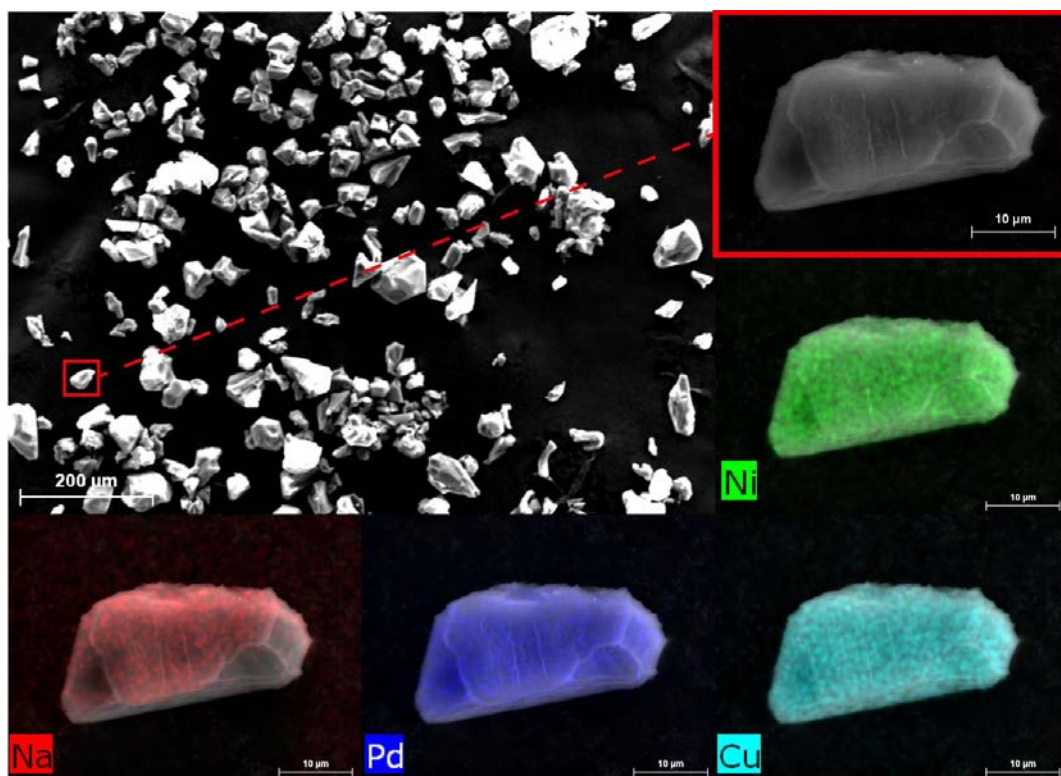
**Supplementary Figure 14.** HR-TEM image of **4**.



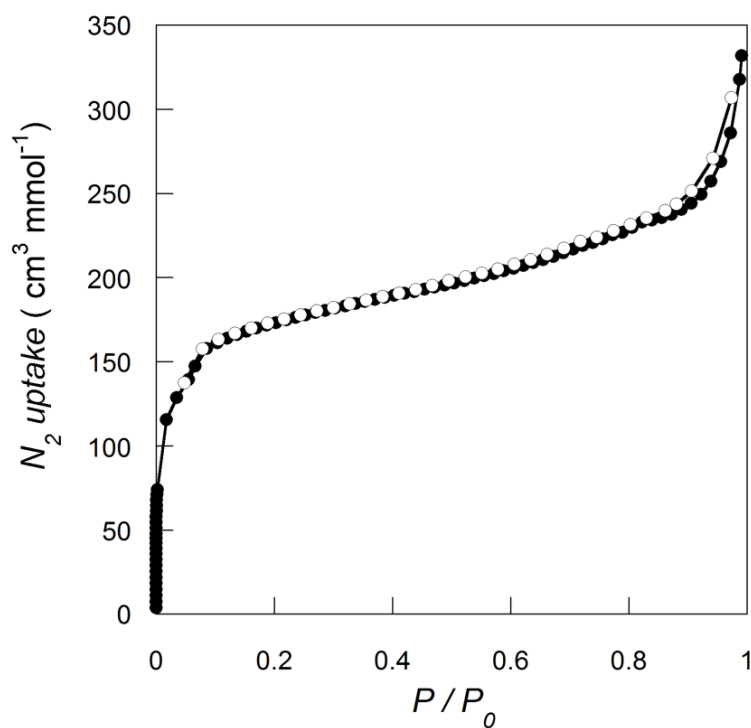
**Supplementary Figure 15.** SEM image of a polycrystalline sample of **4** (a) as well as the corresponding EDX elemental mapping of the bulk for Cu, Ni, Pd, and Na elements (b).



**Supplementary Figure 16.** SEM image of a polycrystalline sample of **4** as well as the corresponding EDX elemental mapping of one selected crystal of **4** for Cu, Ni, Pd, and Na elements.



**Supplementary Figure 17.** SEM image of a polycrystalline sample of **4** as well as the corresponding EDX elemental mapping of one selected crystal of **4** for Cu, Ni, Pd, and Na elements.



**Supplementary Figure 18.**  $N_2$  (77 K) adsorption isotherm for the activated compound **4**. Filled and empty symbols indicate the adsorption and desorption isotherms, respectively. The sample was activated at 80 °C under reduced pressure for 16 h prior to carry out the sorption measurements.

**Preparation of diazocompound 8:** In a 250 ml round-bottomed flask, 3-phenylpropanoyl chloride (1.48 ml, 10 mmol) was dissolved in 50 ml of a 1:1 (v:v) mixture of dry THF:CH<sub>3</sub>CN solvents, and the solution was cooled to 0 °C. Then, a commercial 2M solution of TMSCHN<sub>2</sub> in Et<sub>2</sub>O (10 ml, 20 mmol, 2 equiv.) was added, and the mixture was magnetically stirred at room temperature for 4 h. After aqueous work-up, drying and rotavapory concentration, product **8** was isolated by column chromatography on silica as an orange oil (1.50 g, 86%).

### Catalytic experiments:

Crystals of **3** and **4** give exactly the same catalytic results than the corresponding powdered polycrystalline samples.

*Buchner reaction.* Pd<sub>4</sub> MOF (10 mg, 0.5 mol%) was placed in a 50 ml two-necked round-bottomed flask equipped with a magnetic stirrer and a condenser. Then, the benzene derivative **5** (30 ml) was added and the flask placed in a pre-heated bath oil at 80 °C. A solution of the corresponding diazoacetate **6** (0.75 mmol) in **5** (10 ml) was added during 5 h with a syringe pump, and the reaction was left for additional 3 h. After cooling, the solid catalyst was filtered off, washed with dichloromethane, and reused. The filtrates were concentrated under reduced pressure to give the final products as yellow oils.

For batch experiments, the diazoacetate solution is added at once, from the beginning of the reaction.

For kinetics, aliquots of 1 ml were periodically taken and analyzed by GC-MS after cooling and derivatization to the more stable conjugated isomers with CF<sub>3</sub>COOH or Et<sub>3</sub>N.

For *in-situ* liquid NMR experiments, the reaction was run in deuterated solvents in ten times less scale and analyzed periodically at room temperature.

For reactions with Rh<sub>2</sub>(OAc)<sub>4</sub> catalyst, the procedure was as indicated for Pd<sub>4</sub> MOF.

*Alcohol insertion.* Pd<sub>4</sub> MOF (0.4 mg, 0.005 mol%) was placed in a 25 ml round-bottomed flask equipped with a magnetic stirrer. Then, the corresponding alcohol **11** (5

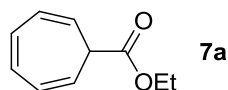
ml) and diazoacetate **6** (2.5 mmol) were added, a condenser was fitted, and the flask placed in a pre-heated bath oil at 80 °C. After 2 h and cooling, the solid catalyst was filtered off, and the filtrates were concentrated under reduced pressure to give the final products as yellow oils.

*Dimerization.* Pd<sub>4</sub> MOF (0.4 mg, 0.005 mol%) was placed in a 25 ml round-bottomed flask equipped with a magnetic stirrer. Then, a solution of the corresponding diazoacetate **6** (2.5 mmol) in toluene (5 ml) was added, a condenser was fitted, and the flask placed in a pre-heated bath oil at 80 °C. After 2 h and cooling, the solid catalyst was filtered off, and the filtrates were concentrated under reduced pressure to give the final products as yellow oils.

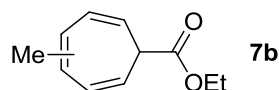
**Buchner reaction in flow:** A paper cartridge with catalyst **4** (100 mg) was placed in a 25 mL Soxhlet apparatus connected on the top with a condenser. Benzene **5a** (100 ml) was placed in a 250 ml two-necked round-bottomed flask equipped with a magnetic stirrer and connected to the Soxhlet apparatus at the bottom, and the flask was placed in a pre-heated bath oil at 80 °C. In this way, **5a** re-circulates all the time through the catalyst. Then, **6a** (10 mmol) was added with a syringe pump under continuously refluxing for 24 h. In this way, a solution of ~1 gram of cycloheptatriene **7a** in 100 ml of benzene (~0.1 M, 90% yield) was obtained. The solid catalyst was recovered and washed with dichloromethane, and reused.

**In-situ magic angle spinning–solid nuclear magnetic resonance (MAS–NMR) experiment:** 50 mg of catalyst **3** were introduced into glass inserts and dehydrated at 80 °C during 18 h. Then, 4.7 μmol of isotopically-labelled **6a** N<sub>2</sub><sup>13</sup>CH–COOEt, corresponding to Pd/**6a**=1, were introduced onto the dehydrated material. The glass inserts were sealed while they were immersed into liquid nitrogen. <sup>13</sup>C Solid-state NMR spectra were recorded at room temperature with a Bruker AVIII HD 400 WB spectrometer. The glass inserts were fitted into 7 mm rotors and were spun at 5 kHz in a Bruker BL7 probe. <sup>13</sup>C CP/MAS NMR spectra were recorded with proton decoupling, with <sup>1</sup>H 90° pulse length of 5 μs, and a recycle delay of 3s.

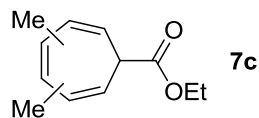
**Product characterization:** All products are yellow oils, prepared as indicated in the general reaction procedure and with satisfactory analysis according to previous literature (see main text, references 28–29).



Ethyl cyclohepta-2,4,6-trienecarboxylate **7a** (91%). IR ( $\nu$ ,  $\text{cm}^{-1}$ ): 2981, 1737.  $^1\text{H}$  NMR ( $\delta$ , ppm;  $J$ , Hz): 6.58 (2xCH, 2H, t,  $J$  = 3.0), 6.19 (2xCH, 2H, d,  $J$  = 8.9), 5.36 (2xCH, 2H, dd,  $J$  = 8.9, 5.6), 4.19 (CH<sub>2</sub>, 2H, q,  $J$  = 7.0), 2.46 (CH, 1H, t,  $J$  = 5.6), 1.24 (CH<sub>3</sub>, 3H, t,  $J$  = 7.1).  $^{13}\text{C}$  NMR ( $\delta$ , ppm): 173.1 (C), 130.9 (2xCH), 125.6 (2xCH), 117.2 (2xCH), 61.0 (CH<sub>2</sub>), 44.1 (CH), 14.2 (CH<sub>3</sub>).

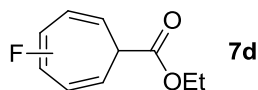


Ethyl methylcyclohepta-2,4,6-trienecarboxylate **7b** (57%, 2-methyl/3-methyl/4-methyl <5:42:53). IR ( $\nu$ ,  $\text{cm}^{-1}$ ): 2927, 1743.  $^1\text{H}$  NMR ( $\delta$ , ppm;  $J$ , Hz; diagnostic signals underlined): 2-methyl: 6.48–5.55 (5xCH, 5H), 4.20 (CH<sub>2</sub>, 2H, q,  $J$  = 7.0), 2.75 (CH, 1H, d,  $J$  = 6.7), 2.19 (CH<sub>3</sub>, 3H, s), 1.26 (CH<sub>3</sub>, 3H, t,  $J$  = 7.0); 3-methyl: 6.39–5.98 (3xCH, 3H), 4.95–4.65 (2xCH, 2H), 4.14 (CH<sub>2</sub>, 2H, q,  $J$  = 7.0), 2.15 (CH, 1H, t,  $J$  = 5.0), 1.86 (CH<sub>3</sub>, 3H, s), 1.21 (CH<sub>3</sub>, 3H, t,  $J$  = 7.0); 4-methyl: 6.39–5.98 (2xCH, 2H), 5.35 (3xCH, 3H, mult), 4.15 (CH<sub>2</sub>, 2H, q,  $J$  = 7.0), 2.47 (CH, 1H, t,  $J$  = 5.3), 1.98 (CH<sub>3</sub>, 3H, s), 1.22 (CH<sub>3</sub>, 3H, t,  $J$  = 7.0).  $^{13}\text{C}$  NMR ( $\delta$ , ppm; unless indicated, indistinguishable signals for each isomer): 173.9–169.7 (C), 139.8 (2xC 2-methyl), 131.2–113.8 (CH), 68.1–60.7 (CH<sub>2</sub>), 48.9–39.7 (CH), 24.0–20.7 (CH<sub>3</sub>), 14.2–14.0 (CH<sub>3</sub>).

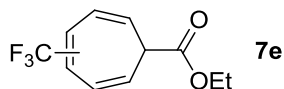


Ethyl dimethylcyclohepta-2,4,6-trienecarboxylate **7c** (64%, 1,4-dimethyl/2,5-dimethyl/3,6-dimethyl <5:26:69). IR ( $\nu$ ,  $\text{cm}^{-1}$ ): 2983, 2931, 1741.  $^1\text{H}$  NMR ( $\delta$ , ppm;  $J$ , Hz; diagnostic signals underlined): 2,5-dimethyl: 6.14–5.59 (2xCH, 2H), 4.71–4.18 (2xCH, 2H), 4.24 (CH<sub>2</sub>, 2H, q,  $J$  = 7.0), 2.79 (CH, 1H, d,  $J$  = 7.0), 1.96 (CH<sub>3</sub>, 3H, s), 1.87 (CH<sub>3</sub>, 3H, s), 1.20 (CH<sub>3</sub>, 3H, t,  $J$  = 7.0); 3,6-dimethyl: 7.17–7.02 (2xCH, 2H), 5.99–5.92 (2xCH, 2H), 4.20 (CH<sub>2</sub>, 2H, q,  $J$  = 7.0), 2.24 (CH, 1H, t,  $J$  = 10.0), 1.88 (2xCH<sub>3</sub>, 6H, s), 1.21 (CH<sub>3</sub>, 3H, t,  $J$  = 7.0).  $^{13}\text{C}$  NMR ( $\delta$ , ppm;

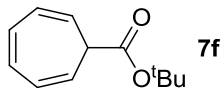
indistinguishable signals for each isomer): 178.7–171.0 (C), 132.8 (2xC 3,6-dimethyl), 128.7–119.2 (CH), 63.6–60.6 (CH<sub>2</sub>), 51.5–47.2 (CH), 23.8–20.4 (CH<sub>3</sub>), 14.2–13.8 (CH<sub>3</sub>).

**7d**

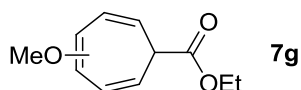
Ethyl fluorocyclohepta-2,4,6-trienecarboxylate **7d** (78%, 2-fluoro/3-fluoro/4-fluoro n.d.:<5:95). IR ( $\nu$ , cm<sup>-1</sup>): 2925, 1741. <sup>1</sup>H NMR ( $\delta$ , ppm; *J*, Hz; 4-fluoro): 6.31 (CH, 1H, ddd, *J* = 16.8, 6.6, 1.8), 6.29 (CH, 1H, mult), 5.61 (CH, 1H, quint, *J* = 5.3), 5.30 (CH, 1H, dd, *J* = 9.4, 5.6), 4.62 (CH, 1H, t, *J* = 5.0), 4.15 (CH<sub>2</sub>, 2H, q, *J* = 7.0), 2.62 (CH, 1H, t, *J* = 5.0), 1.22 (CH<sub>3</sub>, 3H, t, *J* = 7.1). <sup>13</sup>C NMR ( $\delta$ , ppm; *J*<sub>C-F</sub>, Hz; 4-fluoro): 169.7 (C), 162.3 (C, *J* = 253.3), 123.3 (CH, *J* = 10.7), 122.1 (CH, *J* = 13.5), 119.8 (CH, *J* = 34.5), 116.5 (CH, *J* = 2.7), 112.0 (CH, *J* = 23.7), 68.2 (CH<sub>2</sub>), 44.7 (CH), 14.1 (CH<sub>3</sub>). <sup>19</sup>F NMR ( $\delta$ , ppm; *J*, Hz; 4-fluoro): -98.36.

**7e**

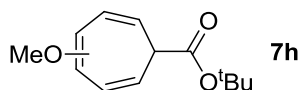
Ethyl trifluoromethylcyclohepta-2,4,6-trienecarboxylate **7e** (48%, 2-trifluoromethyl/3-trifluoromethyl /4-trifluoromethyl <5:42:58). IR ( $\nu$ , cm<sup>-1</sup>): 2924, 1742. <sup>1</sup>H NMR ( $\delta$ , ppm; *J*, Hz; diagnostic signals underlined): 3-trifluoromethyl: 6.85–6.17 (2xCH, 2H), 4.74–4.42 (3xCH, 3H), 4.17 (CH<sub>2</sub>, 2H, q, *J* = 7.0), 1.96 (CH, 1H, t, *J* = 6.5), 1.20 (CH<sub>3</sub>, 3H, t, *J* = 7.0); 4-trifluoromethyl: 6.95–6.23 (3xCH, 3H), 5.21–5.78 (2xCH, 2H), 4.20 (CH<sub>2</sub>, 2H, q, *J* = 7.0), 2.38 (CH, 1H, t, *J* = 3.0), 1.24 (CH<sub>3</sub>, 3H, t, *J* = 7.0). <sup>13</sup>C NMR ( $\delta$ , ppm; indistinguishable signals for each isomer): 171.5–165.0 (C), 133.6–114.6 (CH and CF<sub>3</sub>), 63.5–61.0 (CH<sub>2</sub>), 31.4–30.2 (CH), 14.1–13.8 (CH<sub>3</sub>). <sup>19</sup>F NMR ( $\delta$ , ppm; *J*, Hz): -64.99, -65.26.

**7f**

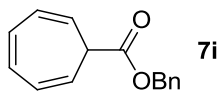
*tert*-Butyl cyclohepta-2,4,6-trienecarboxylate **7f** (32%). IR ( $\nu$ , cm<sup>-1</sup>): 2926, 1732. <sup>1</sup>H NMR ( $\delta$ , ppm; *J*, Hz): 6.56 (2xCH, 2H, t, *J* = 4.0), 6.16 (2xCH, 2H, d, *J* = 8.9), 5.34 (2xCH, 2H, dd, *J* = 8.9, 5.6), 2.38 (CH, 1H, t, *J* = 5.6), 1.48 (3xCH<sub>3</sub>, 9H, s). <sup>13</sup>C NMR ( $\delta$ , ppm): 172.2 (C), 134.5 (2xCH), 128.3 (2xCH), 118.4 (2xCH), 45.1 (CH), 27.9 (3xCH<sub>3</sub>).

Ethyl methoxycyclohepta-2,4,6-trienecarboxylate **7g**

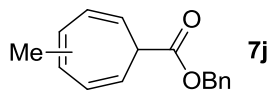
(92%, 2-methoxy/3-methoxy/4-methoxy 27:<5:68). IR ( $\nu$ ,  $\text{cm}^{-1}$ ): 2984, 1738.  $^1\text{H}$  NMR ( $\delta$ , ppm;  $J$ , Hz; diagnostic signals underlined): 2-methoxy: 6.42–6.16 (3xCH, 3H), 5.48–5.40 (2xCH, 2H), 4.11 (CH<sub>2</sub>, 2H, q,  $J$  = 7.0), 3.56 (CH<sub>3</sub>, 3H, s), 3.23 (CH, 1H, d,  $J$  = 7.3), 1.18 (CH<sub>3</sub>, 3H, t,  $J$  = 7.1); 4-methoxy: 6.13 (CH, 1H, ddd,  $J$  = 8.0, 6.7, 1.1), 6.00 (CH, 1H, dt,  $J$  = 9.9, 3.3), 5.79 (CH, 1H, dd,  $J$  = 6.6, 1.7), 5.54 (CH, 1H, quintd,  $J$  = 5.6, 1.7), 5.19 (CH, 1H, ddd,  $J$  = 8.8, 5.2, 0.7), 4.18 (CH<sub>2</sub>, 2H, q,  $J$  = 7.0), 3.58 (CH<sub>3</sub>, 3H, s), 2.62 (CH, 1H, t,  $J$  = 5.0), 1.23 (CH<sub>3</sub>, 3H, t,  $J$  = 7.1).  $^{13}\text{C}$  NMR ( $\delta$ , ppm): 2-methoxy: 170.1 (C), 149.1 (C), 128.0 (CH), 126.9 (CH), 124.5 (CH), 117.8 (CH), 97.3 (CH), 60.9 (CH<sub>2</sub>), 56.6 (CH<sub>3</sub>), 48.9 (CH), 14.3 (CH<sub>3</sub>); 4-methoxy: 172.9 (C), 160.0 (C), 124.8 (CH), 122.1 (CH), 119.9 (CH), 117.1 (CH), 104.8 (CH), 61.0 (CH<sub>2</sub>), 54.7 (CH<sub>3</sub>), 44.0 (CH), 14.2 (CH<sub>3</sub>).

*tert*-Butyl methoxycyclohepta-2,4,6-trienecarboxylate **7h**

(46%, 2-methoxy/3-methoxy/4-methoxy 15:<5:85). IR ( $\nu$ ,  $\text{cm}^{-1}$ ): 2979, 1732.  $^1\text{H}$  NMR ( $\delta$ , ppm;  $J$ , Hz; diagnostic signals underlined; 4-methoxy): 6.12 (CH, 1H, 7,  $J$  = 8.8), 5.98 (CH, 1H, d,  $J$  = 8.8), 5.78 (CH, 1H, d,  $J$  = 6.2), 5.53 (CH, 1H, dd,  $J$  = 9.3, 5.4), 5.19 (CH, 1H, dd,  $J$  = 8.8, 5.4), 3.58 (CH<sub>3</sub>, 3H, s), 2.55 (CH, 1H, t,  $J$  = 5.0), 1.43 (3xCH<sub>3</sub>, 9H, s).  $^{13}\text{C}$  NMR ( $\delta$ , ppm): 4-methoxy: 174.0 (C), 160.0 (C), 124.5 (CH), 122.0 (CH), 121.1 (CH), 114.4 (CH), 104.7 (CH), 54.7 (CH<sub>3</sub>), 45.1 (CH), 28.1 (3xCH<sub>3</sub>).

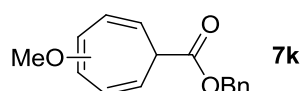
Benzyl cyclohepta-2,4,6-trienecarboxylate **7i** (73%).

IR ( $\nu$ ,  $\text{cm}^{-1}$ ): 2985, 1738.  $^1\text{H}$  NMR ( $\delta$ , ppm;  $J$ , Hz): 7.43–7.34 (5xCH, 5H), 6.69 (2xCH, 2H, t,  $J$  = 3.1), 6.30 (2xCH, 2H, d,  $J$  = 8.8), 5.50 (2xCH, 2H, dd,  $J$  = 8.8, 5.6), 5.29 (CH<sub>2</sub>, 2H, s), 2.67 (CH, 1H, t,  $J$  = 5.6).  $^{13}\text{C}$  NMR ( $\delta$ , ppm): 172.9 (C), 135.9 (C), 130.9 (2xCH), 128.5 (2xCH), 128.4 (CH), 128.2 (2xCH), 125.7 (2xCH), 116.7 (2xCH), 66.8 (CH<sub>2</sub>), 44.0 (CH).

Benzyl methylcyclohepta-2,4,6-trienecarboxylate **7j**

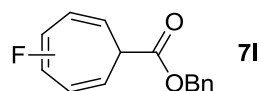
(70%, 2-methyl/3-methyl /4-methyl <5:60:35). IR ( $\nu$ ,  $\text{cm}^{-1}$ ): 3031, 2955, 2924, 1738.

$^1\text{H}$  NMR ( $\delta$ , ppm;  $J$ , Hz; diagnostic signals underlined): *2-methyl*: 7.30–7.10 (5xCH, 5H), 6.42–4.80 (5xCH, 5H), 4.92 (CH<sub>2</sub>, 2H, s), 2.84 (CH, 1H, d,  $J=6.2$ ), 1.84 (CH<sub>3</sub>, 3H, s); *3-methyl*: 7.30–7.10 (5xCH, 5H), 6.45–5.75 (3xCH, 3H), 5.03 (CH<sub>2</sub>, 2H, s), 4.70–4.08 (2xCH, 2H), 2.21 (CH, 1H, t,  $J=8.6$ ), 1.84 (CH<sub>3</sub>, 3H, s); *4-methyl*: 6.45–5.75 (2xCH, 2H), 5.35–4.92 (3xCH, 3H, mult), 5.11 (CH<sub>2</sub>, 2H, s), 2.52 (CH, 1H, t,  $J=6.0$ ), 1.95 (CH<sub>3</sub>, 3H, s).  $^{13}\text{C}$  NMR ( $\delta$ , ppm; unless indicated, indistinguishable signals for each isomer): 173.2–164.7 (C), 139.8–134.3 (C), 133.8–113.2 (CH), 68.2–60.7 (CH<sub>2</sub>), 43.4 (CH, *4-methyl*), 43.0 (CH, *2-methyl*), 39.3 (CH, *3-methyl*), 24.3 (CH<sub>3</sub>, *2-methyl*), 24.0 (CH<sub>3</sub>, *4-methyl*), 21.8 (CH<sub>3</sub>, *3-methyl*).



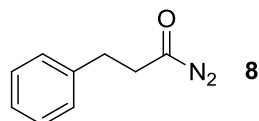
Benzyl methoxycyclohepta-2,4,6-trienecarboxylate **7k**

(95%, *2-methoxy/3-methoxy/4-methoxy* n.d.: n.d.:100). IR ( $\nu$ , cm<sup>-1</sup>): 3032, 2955, 2925, 1736.  $^1\text{H}$  NMR ( $\delta$ , ppm;  $J$ , Hz; diagnostic signals underlined, *4-methoxy*): 7.36–7.19 (5xCH, 5H), 6.16 (CH, 1H, q,  $J=8.0$ ), 6.07 (CH, 1H, d,  $J=8.8$ ), 5.77 (CH, 1H, d,  $J=6.2$ ), 5.52 (CH, 1H, dd,  $J=8.8, 6.2$ ), 5.18 (CH, 1H, dd,  $J=8.4, 5.1$ ), 5.11 (CH<sub>2</sub>, 2H, s), 3.53 (CH<sub>3</sub>, 3H, s), 2.66 (CH, 1H, t,  $J=5.3$ ).  $^{13}\text{C}$  NMR ( $\delta$ , ppm, *4-methoxy*): 172.8 (C), 160.0 (C), 135.9 (C), 128.7 (CH), 128.6 (2xCH), 128.1 (2xCH), 125.0 (CH), 122.1 (CH), 119.4 (CH), 112.8 (CH), 105.0 (CH), 66.8 (CH<sub>2</sub>), 54.7 (CH<sub>3</sub>), 44.0 (CH).



Benzyl fluorocyclohepta-2,4,6-trienecarboxylate **7l** (65%,

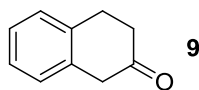
*2-fluoro/3-fluoro/4-fluoro* n.d.:n.d.:100). IR ( $\nu$ , cm<sup>-1</sup>): 3032, 2956, 1739.  $^1\text{H}$  NMR ( $\delta$ , ppm;  $J$ , Hz; *4-fluoro*): 7.28–7.05 (5xCH, 5H), 6.40 (CH, 1H, t,  $J=4.8$ ), 6.11 (CH, 1H, d,  $J=7.4$ ), 5.58 (CH, 1H, quint,  $J=5.4$ ), 5.28 (CH, 1H, dd,  $J=8.0, 5.7$ ), 5.12 (CH<sub>2</sub>, 2H, s), 4.66 (CH, 1H, t,  $J=11.0$ ), 2.65 (CH, 1H, t,  $J=5.1$ ).  $^{13}\text{C}$  NMR ( $\delta$ , ppm;  $J_{\text{C-F}}$ , Hz; *4-fluoro*): 167.3 (C,  $J=349.0$ ), 164.7 (C), 135.3 (C), 133.8 (CH), 128.7 (2xCH), 128.1 (2xCH), 123.4 (CH,  $J=10.8$ ), 122.0 (CH,  $J=13.4$ ), 121.9 (CH,  $J=2.5$ ), 118.6 (CH,  $J=7.5$ ), 112.1 (CH,  $J=25.4$ ), 67.1 (CH<sub>2</sub>), 43.0 (CH).



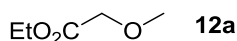
1-Diazo-4-phenylbutan-2-one **8**. Rf (25% AcOEt in *n*-

hexane)= 0.52.  $^1\text{H}$  NMR ( $\delta$ , ppm;  $J$ , Hz): 7.39 (2xCH, 2H, t,  $J=6.8$ ), 7.34–7.27 (3xCH,

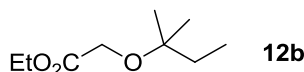
3H, mult), 5.39 (CH, 1H, bs), 3.06 (CH<sub>2</sub>, 2H, t, *J* = 7.5), 2.73 (CH<sub>2</sub>, 2H, t, *J* = 7.0). <sup>13</sup>C NMR (δ, ppm): 195.6 (C), 140.4 (C), 128.2 (2xCH), 128.0 (2xCH), 125.9 (CH), 59.9 (CH), 54.1 (CH<sub>2</sub>), 30.3 (CH<sub>2</sub>).

**9**

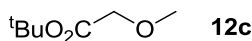
3,4-Dihydronaphthalen-2(1H)-one **9**. <sup>1</sup>H NMR (δ, ppm; *J*, Hz): 6.31 (2xCH, 2H, mult), 6.04 (2xCH, 2H, mult), 4.92 (2xCH, 2H, dd, *J* = 9.4, 3.8), 4.04 (CH, 1H, d, *J* = 4.7), 2.50 (CH<sub>2</sub>, 2H, mult), 2.38 (CH<sub>2</sub>, 2H, mult). <sup>13</sup>C NMR (δ, ppm): 195.6 (C), 140.4 (C), 128.2 (2xCH), 128.0 (2xCH), 125.9 (CH), 59.9 (CH), 54.1 (CH<sub>2</sub>), 30.3 (CH<sub>2</sub>).

**12a**

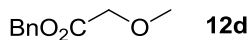
Ethyl 2-methoxyacetate **12a**. IR (ν, cm<sup>-1</sup>): 2982, 1729. <sup>1</sup>H NMR (δ, ppm; *J*, Hz): 4.16 (CH<sub>2</sub>, 2H, mult), 3.96 (CH<sub>2</sub>, 2H, t, *J* = 2.7), 3.38 (CH<sub>3</sub>, 3H, t, *J* = 2.8), 1.28 (CH<sub>3</sub>, 3H, tt, *J* = 7.0, 2.7). <sup>13</sup>C NMR (δ, ppm): 170.1 (C), 69.7 (CH<sub>2</sub>), 60.7 (CH<sub>2</sub>), 59.1 (CH<sub>3</sub>), 14.0 (CH<sub>3</sub>).

**12b**

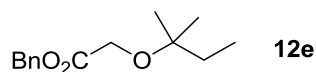
Ethyl 2-(*tert*-pentyloxy)acetate **12b**. IR (ν, cm<sup>-1</sup>): 2976, 2937, 1759, 1731. <sup>1</sup>H NMR (δ, ppm; *J*, Hz): 4.15 (CH<sub>2</sub>, 2H, qt, *J* = 7.1, 3.6), 3.93 (CH<sub>2</sub>, 2H, t, *J* = 3.7), 1.48 (CH<sub>2</sub>, 2H, qt, *J* = 7.4, 2.8), 1.23 (CH<sub>3</sub>, 3H, tt, *J* = 7.0, 3.5), 1.11 (2xCH<sub>3</sub>, 6H, t, *J* = 4.7), 0.85 (CH<sub>3</sub>, 3H, tt, *J* = 7.6, 3.4). <sup>13</sup>C NMR (δ, ppm): 171.2 (C), 76.5 (C), 60.6 (CH<sub>2</sub>), 60.4 (CH<sub>2</sub>), 32.6 (CH<sub>2</sub>), 24.6 (2xCH<sub>3</sub>), 13.9 (CH<sub>3</sub>), 8.1 (CH<sub>3</sub>).

**12c**

*tert*-Butyl 2-methoxyacetate **12c**. IR (ν, cm<sup>-1</sup>): 3004, 2982, 2934, 1735. <sup>1</sup>H NMR (δ, ppm; *J*, Hz): 3.85 (CH<sub>2</sub>, 2H, mult), 3.38 (CH<sub>3</sub>, 3H, t, *J* = 4.7), 1.42 (3xCH<sub>3</sub>, 9H, bs). <sup>13</sup>C NMR (δ, ppm): 169.5 (C), 81.6 (C), 70.2 (CH<sub>2</sub>), 59.0 (CH<sub>3</sub>), 27.9 (3xCH<sub>3</sub>).

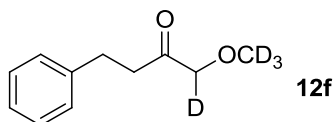
**12d**

Benzyl 2-methoxyacetate **12d**. IR (ν, cm<sup>-1</sup>): 3032, 2940, 1896, 1827, 1755. <sup>1</sup>H NMR (δ, ppm; *J*, Hz): 7.39–7.36 (5xCH, 5H, mult), 5.21 (CH<sub>2</sub>, 2H, s), 4.08 (CH<sub>2</sub>, 2H, s), 3.45 (CH<sub>3</sub>, 3H, bs). <sup>13</sup>C NMR (δ, ppm): 170.1 (C), 135.4 (C), 128.6 (2xCH), 128.4 (2xCH), 126.9 (CH), 69.8 (CH<sub>2</sub>), 66.5 (CH<sub>2</sub>), 59.4 (CH<sub>3</sub>).



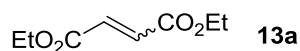
Benzyl 2-(*tert*-pentyloxy)acetate **12e**. IR ( $\nu$ ,  $\text{cm}^{-1}$ ):

3113, 2953, 1695.  $^1\text{H}$  NMR ( $\delta$ , ppm;  $J$ , Hz): 7.38 (5xCH, 5H, mult), 5.23 (CH<sub>2</sub>, 2H, s), 4.81 (CH<sub>2</sub>, 2H, s), 1.53 (CH<sub>2</sub>, 2H, q,  $J$  = 7.5), 1.22 (2xCH<sub>3</sub>, 6H, s), 0.95 (CH<sub>3</sub>, 3H, s).  $^{13}\text{C}$  NMR ( $\delta$ , ppm): 166.7 (C), 135.9 (C), 128.6 (2xCH), 128.4 (2xCH), 126.9 (CH), 71.1 (C), 67.1 (CH<sub>2</sub>), 60.7 (CH<sub>2</sub>), 36.3 (CH<sub>2</sub>), 28.6 (2xCH<sub>3</sub>), 8.6 (CH<sub>3</sub>).



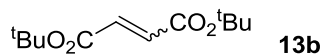
1(*D*)-1-(*D*<sub>3</sub>)Methoxy-4-phenylbutan-2-one **12f**.

$^1\text{H}$  NMR ( $\delta$ , ppm;  $J$ , Hz): 7.18–7.04 (5xCH, 5H, mult), 4.74 (CH, 1H, s), 2.77 (CH<sub>2</sub>, 2H, td,  $J$  = 6.7, 2.0), 2.64 (CH<sub>2</sub>, 2H, td,  $J$  = 6.8, 1.3).  $^{13}\text{C}$  NMR ( $\delta$ , ppm): 210.1 (C), 142.3 (C), 129.4 (2xCH), 129.3 (2xCH), 127.1 (CH), 49.9 (CH), 41.1 (CH<sub>2</sub>), 30.3 (CH<sub>2</sub>).



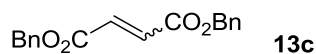
Diethyl but-2-enedioate **13a** (*E/Z* 50:50). IR ( $\nu$ ,  $\text{cm}^{-1}$ ):

2984, 1726, 1645.  $^1\text{H}$  NMR ( $\delta$ , ppm;  $J$ , Hz): 6.83 (2xCH, 2H, s, *Z*), 6.21 (2xCH, 2H, s, *E*), 4.23 (4xCH<sub>2</sub>, 8H, mult), 1.28 (4xCH<sub>3</sub>, 12H, mult).  $^{13}\text{C}$  NMR ( $\delta$ , ppm): 165.0 (2xC), 164.9 (2xC), 133.6 (2xCH), 129.8 (2xCH), 61.8 (4xCH<sub>2</sub>), 14.0 (4xCH<sub>3</sub>).



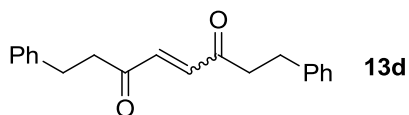
Di-*tert*-butyl but-2-enedioate **13b** (*E/Z* 50:50). IR ( $\nu$ ,  $\text{cm}^{-1}$ ):

2982, 1742, 1691, 1622.  $^1\text{H}$  NMR ( $\delta$ , ppm;  $J$ , Hz): 4.54 (4xCH, 4H, s, *Z+E*), 1.40 (12xCH<sub>3</sub>, 36H, s).  $^{13}\text{C}$  NMR ( $\delta$ , ppm): 166.2 (4xC), 129.0 (2xCH), 128.2 (2xCH), 81.4 (4C), 27.9 (12xCH<sub>3</sub>).



Dibenzyl but-2-enedioate **13c** (*E/Z* 65:35). IR ( $\nu$ ,  $\text{cm}^{-1}$ ):

3111, 3036, 2956, 1745, 1694.  $^1\text{H}$  NMR ( $\delta$ , ppm;  $J$ , Hz): 7.44–7.38 (20xCH, 20H, mult, *Z+E*), 5.24 (4xCH<sub>2</sub>, 8H, bs, *Z+E*), 4.82 (2xCH, 2H, s, *Z*), 4.72 (2xCH, 2H, s, *E*).  $^{13}\text{C}$  NMR ( $\delta$ , ppm): 166.6 (4xC), 135.8 (4xC), 129.0 (4xCH), 128.6 (8xCH), 128.4 (8xCH), 126.9 (2xCH), 125.3 (2xCH), 65.2 (4xCH<sub>2</sub>).



1,8-Diphenyloct-4-ene-3,6-dione **13d**

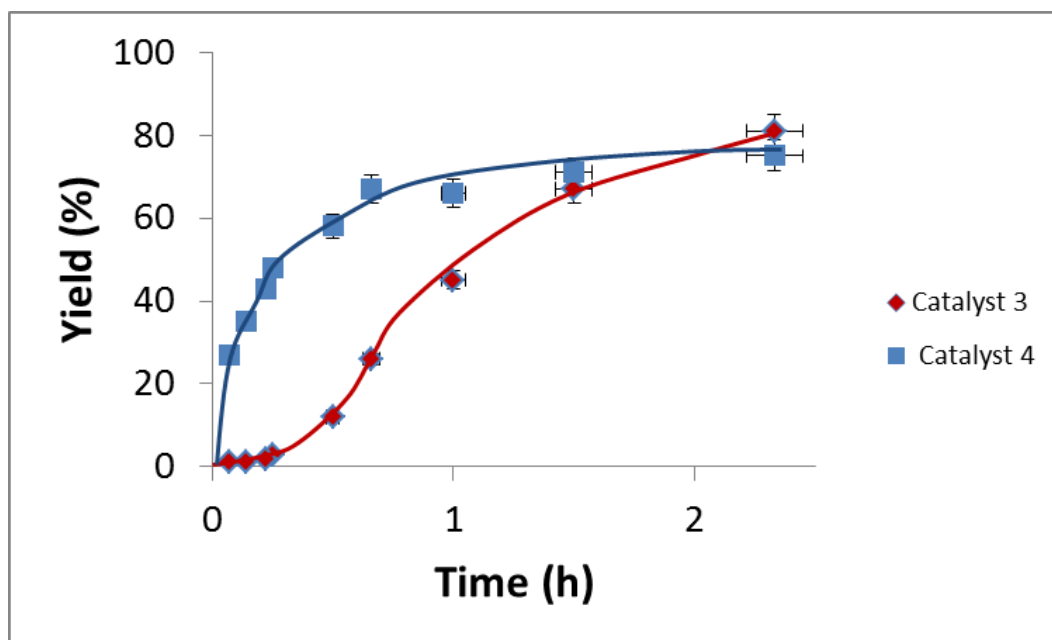
(*E/Z* 50:50).  $^1\text{H}$  NMR ( $\delta$ , ppm;  $J$ , Hz): 7.24–7.08 (20xCH, 20H, mult, *Z+E*), 6.75

(4xCH, 4H, s, *Z+E*), 2.87 (8xCH<sub>2</sub>, 16H, bs, *Z+E*). <sup>13</sup>C NMR (δ, ppm): 199.4 (4xC), 140.4 (4xC), 136.3 (4xCH), 128.6 (8xCH), 128.3 (8xCH), 126.3 (4xCH), 43.1 (4xCH<sub>2</sub>), 29.6 (4xCH<sub>2</sub>).

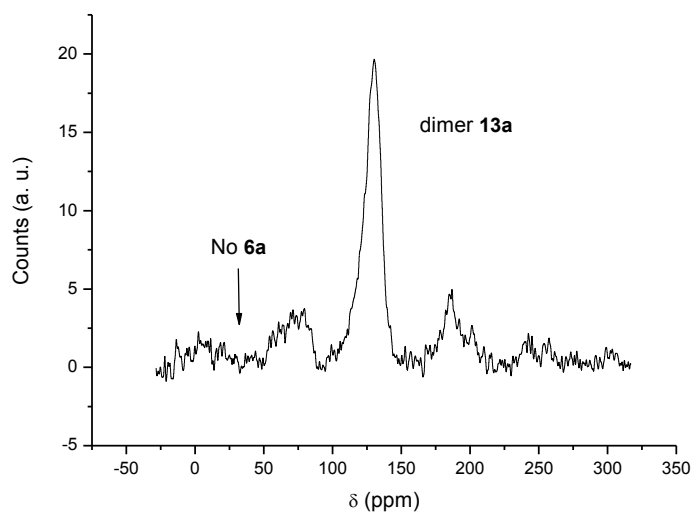
**Table 3.** Isomeric ratio of the intermolecular Buchner reaction with **4** or Rh<sub>2</sub>(OAc)<sub>4</sub> as a catalyst.

| Product   | Isomeric ratio                          | Catalyst <b>4</b> | Rh <sub>2</sub> (OAc) <sub>4</sub>                          |
|-----------|-----------------------------------------|-------------------|-------------------------------------------------------------|
| <b>7b</b> | 2-methyl/3-methyl/4-methyl              | <5:42:53          | 17:23:56<br>11:17:72 <sup>a</sup><br>5:10:85 <sup>a,b</sup> |
| <b>7c</b> | 1,4-dimethyl/2,5-dimethyl /3,6-dimethyl | <5:26:69          |                                                             |
| <b>7d</b> | 2-fluoro/3-fluoro/4-fluoro              | n.d.:<5:95        | n.d.:<5:95<br>8:12:80 <sup>a,b</sup>                        |
| <b>7g</b> | 2-methoxy/3-methoxy/4-methoxy           | 27:<5:68          | 19:<5:76<br>31:9:60 <sup>a,b</sup>                          |

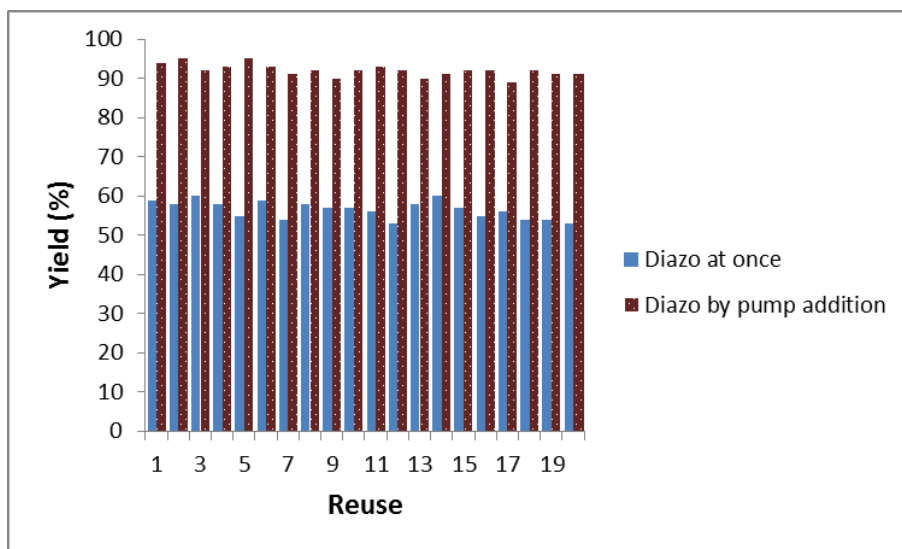
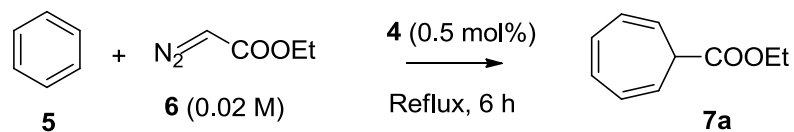
<sup>a</sup> With Rh(II)trifluoroacetate dimer, taken from reference 47 in the main text. <sup>b</sup> With methyldiazoacetate.

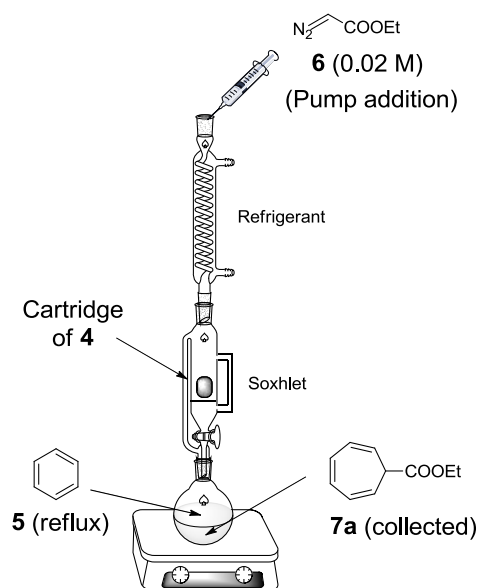


**Supplementary Figure 19.** Kinetics for the intermolecular reaction of **5** and **6** added at once (0.02 M) with **3** or **4** (1 mol%) as catalysts. Average of two measurements. Error bars account for 5% uncertainty.

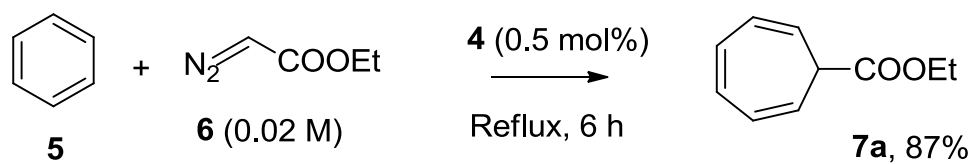


**Supplementary Figure 20.** In-situ magic angle spinning–solid nuclear magnetic resonance (MAS–NMR) measurements of **3** treated with isotopically-labelled **6**  $\text{N}_2^{13}\text{CH-COOEt}$ .

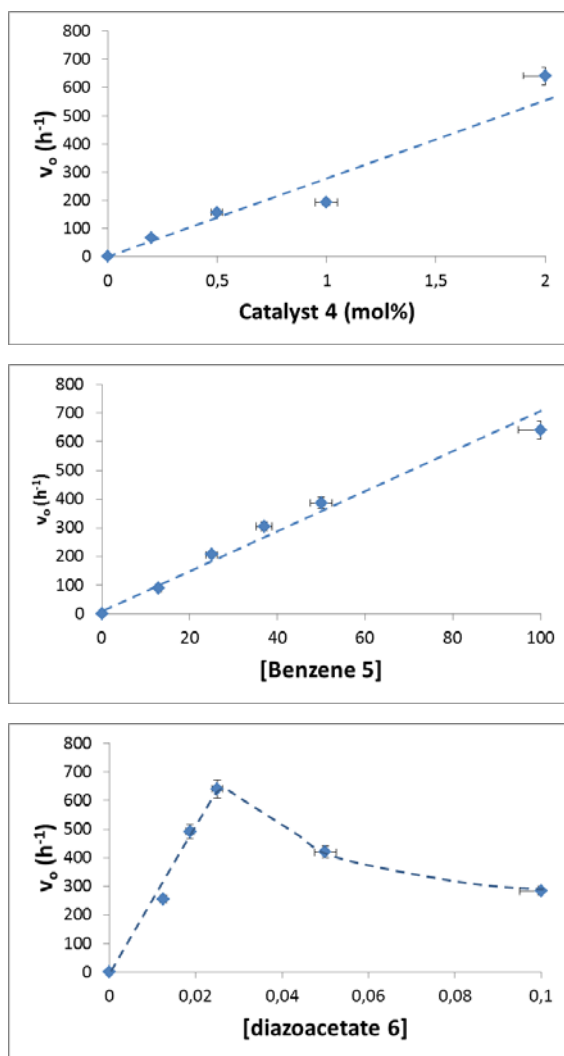
Supplementary Figure 21. Reuses of **4**.



**Supplementary Figure 22.** Reaction in flow with **4**.



**Supplementary Figure 23.** Reaction with **4** after use in flow.



**Supplementary Figure 24.** Determination of the reaction orders for each reactant of the intermolecular Buchner reaction catalysed by **4**. Error bars account for 5% uncertainty.